

CAEfatigue and Robust Design



Increasing product quality, durability,
reducing recalls and experimental testing

Hexagon Manufacturing Intelligence
Jan 2024

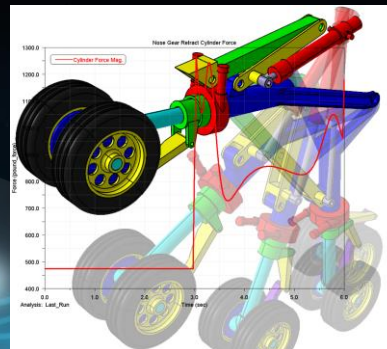
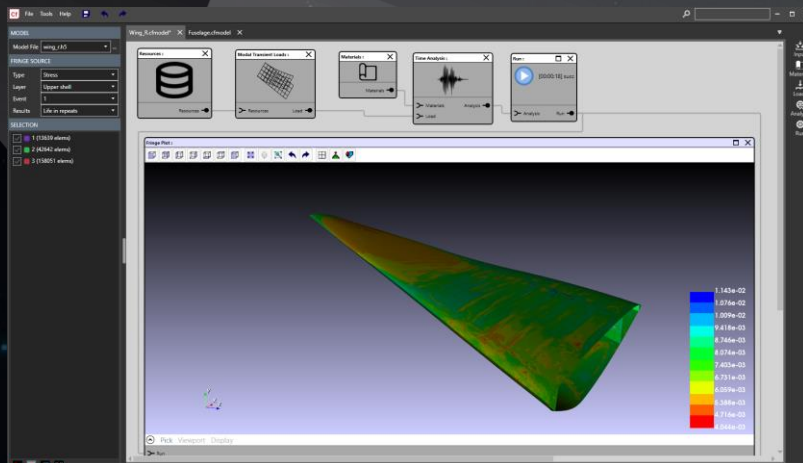
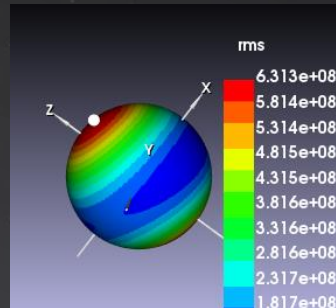
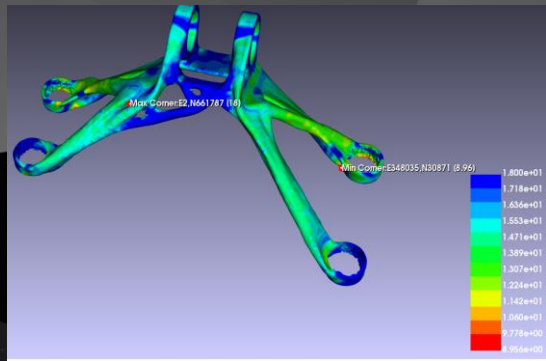
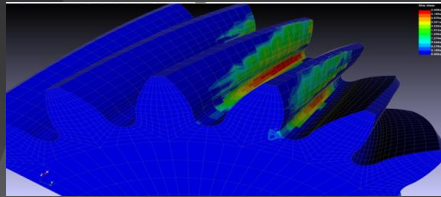


Marco Veltri, PhD

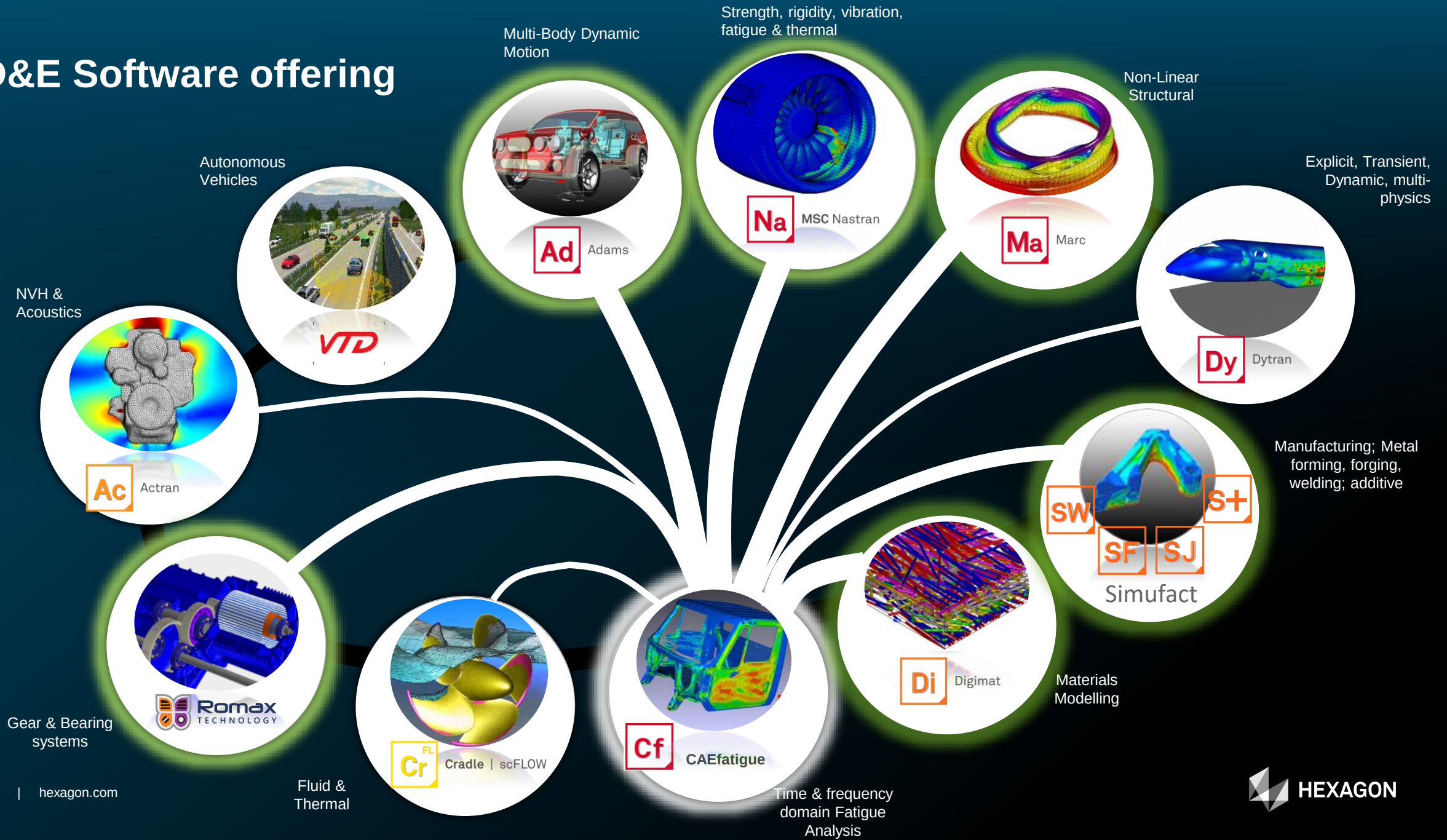
Business Enablement Manager, Multiphysics

Product Management & Business Development

marco.veltri@hexagon.com



D&E Software offering



Fatigue, Loads Management Robust Design : Where are your challenges?

Better
Products
Quicker

Time, Budget,
Quality &
Repeatability
important for a
project delivery
organisation

new platforms
– More
complex
engineering,
more levels of
integration

Digital
handshake
between
engineering
toolsets

Digital
handshake
between
engineering and
operations

Product quality
and robustness

Insights into
complex
systems
limited

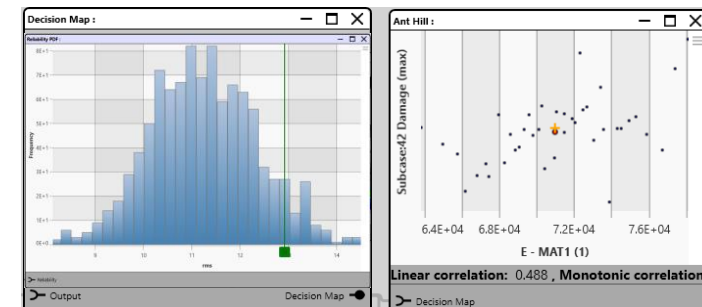
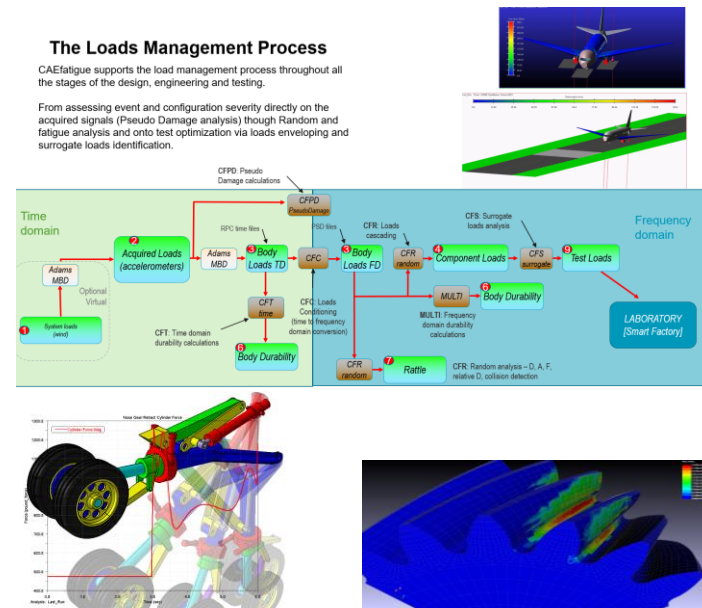
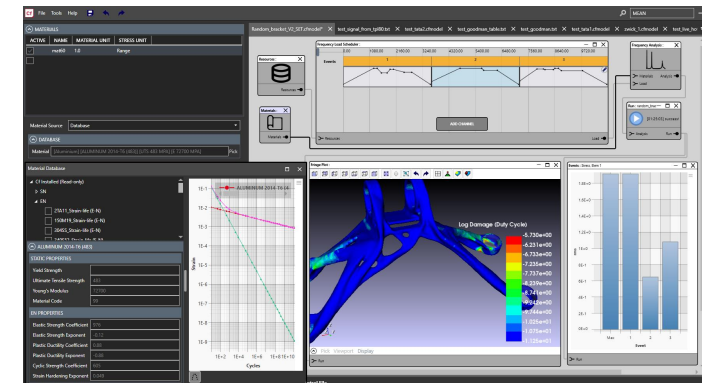
... what we have heard or experienced based from Customer challenges

- Time and Quantity: Solution Speed, Repeatability, Ease of Use, Support (from self-help tutorials to multidomain and Multiphysics solution experts)
- Complexity: Skill-set requirements in challenging domain (dynamics, fatigue, random signals, stochastics...), embedding / accessing company and experts knowhow, integration between CAE, testing, manufacturing processes
- Digital Handshake: operations loads management (e.g. proving ground or MBD), handling certification standards in simulations or generating surrogate loads for physical tests
- Design Quality/Robustness : handling uncertainties and variability, reduce liability, improve correlation with physical world, define cost effective quality checks and tolerances

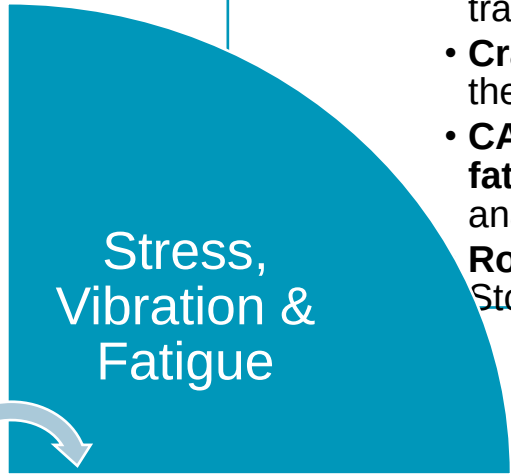
...From challenges to solutions

A quick glance at CAEfatigue & Robust Design value *what makes it unique*

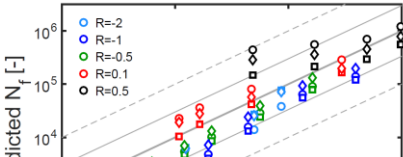
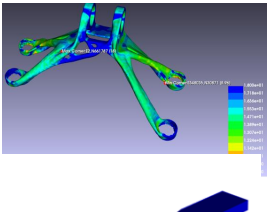
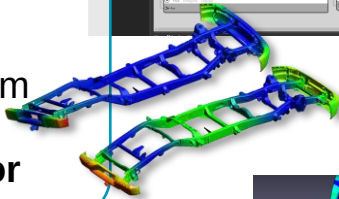
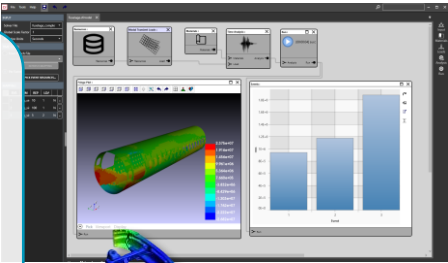
- CAEfatigue architecture
 - Fast and intuitive Process Flow templates
 - Solver neutral (Nastran, Abaqus, Ansys...)
 - batch runs and easy integration in enterprise portals (e.g. SimManager)
 - Visual Random and Vibration response tools
- Load Management Process
 - Digital Signal Processing and automatic conditioning for time to PSD loading conversion
 - Using CAEfatigue for test optimization (envelopes and surrogates)
 - Virtual Strain gauge analysis
- Fatigue analysis integration in CAE processes
 - Embedded fatigue solutions, Manufacturing effects, mechanism and transmission durability, advanced materials
- Robust Design for stochastic simulation and stochastic design improvements



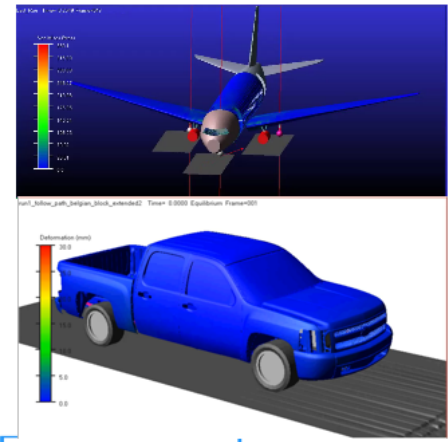
CAEfatigue and Robust Design for Loads Management, Durability Solutions Integration (digital handshake)



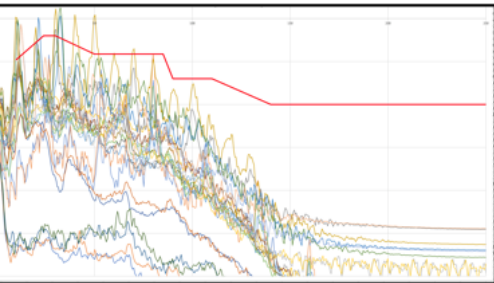
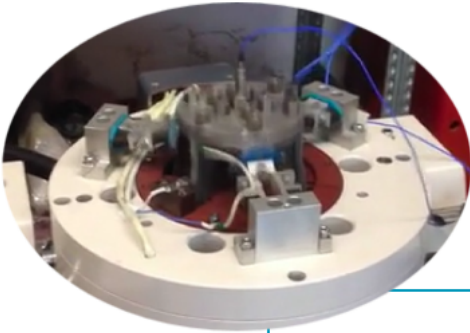
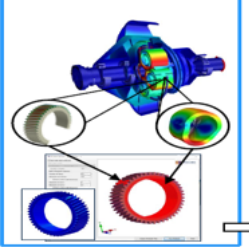
- **Apex/ Nastran** for modal reduction and transfer function
 - **Cradle/Marc** for thermal-structural
 - **CAEfatigue** for **fatigue** and random analysis
- Robust Design for Stochastic Analysis**



Loads Management & Durability Solutions Integration and digital handshake



- **Adams/Apex** for loads in Dynamic transient
- **Romax** for engine, transmission loads, gear/bearing fatigue
- **CAEfatigue** Pseudo damage, load conditioning



- **CAEfatigue** for loads specification and test optimization
- **Robust Design** for test correlation

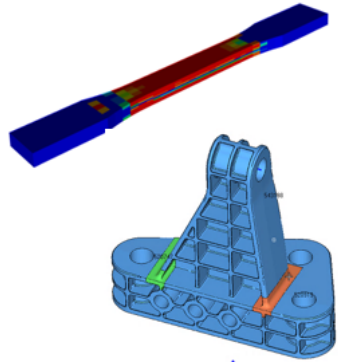
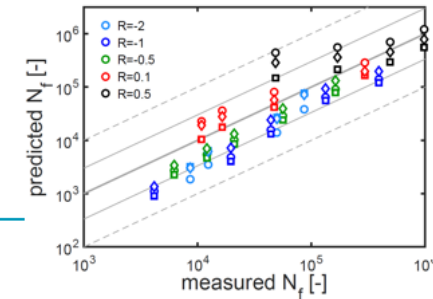
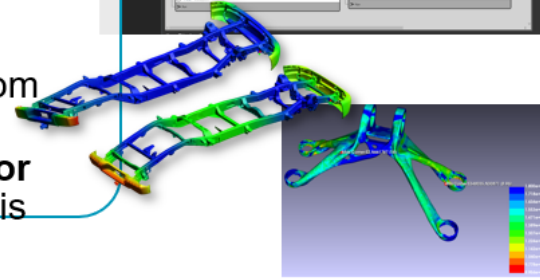
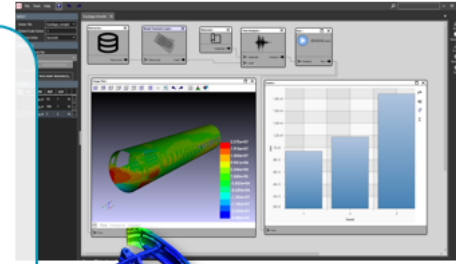
Vehicle
Dynamics &
Durability

Stress,
Vibration &
Fatigue

Testing &
Certification

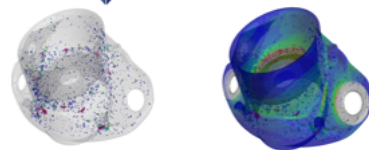
Materials &
Manufacturing
effects

- **Apex/ Nastran** for modal reduction and transfer function
- **Cradle/Marc** for thermal-structural
- **CAEfatigue** for fatigue and random analysis
- **Robust Design for Stochastic Analysis**



- **Digimat** for material engineering
- **Simufact** for seam welds, cold forming processes, AM
- **Volume Graphics CT scan analysis** for defect identification

**VOLUME
GRAPHICS**



HEXAGON

CAEfatigue technologies and operation modes

*Includes Robust Design
(New in 2022)*

**Cf
TIME**

**Cf
RANDOM**

**Cf
FREQUENCY**

**Cf
PREMIUM**

**FE Solver
Embedded
(Nastran/Marc)**

Control File

**Stand Alone
(APEX Aligned)**

Supports Nastran, ANSYS,
Optistruct, Abaqus, Marc, LS
DYNA

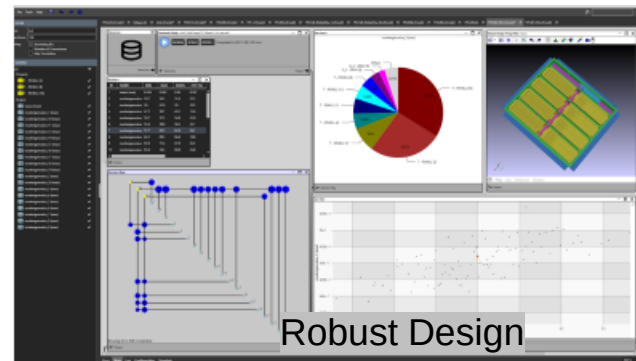
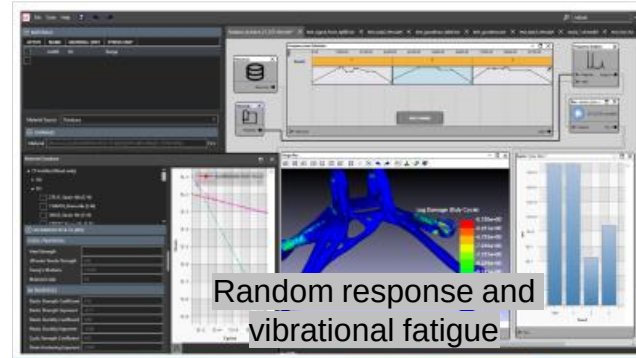
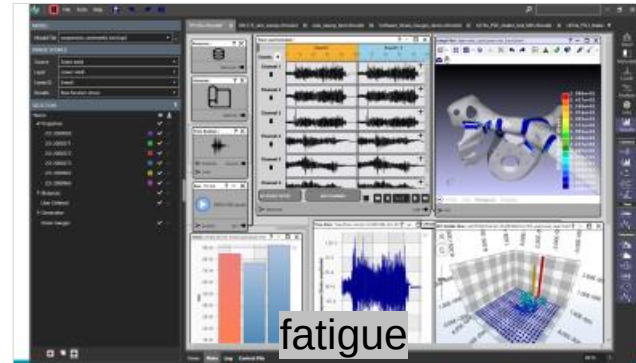
One unified Durability, Random Response and Loads Management solution

CAEfatigue & Robust Design: intuitive, repeatable, transferable process flows

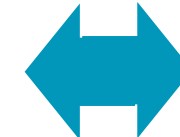
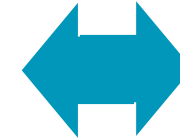
- Process orientated
- Pre-set templates, customizable
- Drag and drop
- Indexed help
- Multi tab / multi output / Multi run

Multiple material sources and models

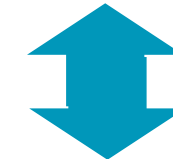
- own or custom DB
- S-N and Strain-Life
- Digimat (SFRP)



Bidirectional flow with batch operatable control file



```
$ Generated by CAEfatigue (2023.1 7114)
$
vibfat 777 csv nash5 corner 1 Digi_Marc_H5_test
$ @matid[60]="mat1"
vmatftg 60 MPa range
file 60 sn ID=Stress-life (S-N) ParentID=2014_HV_T6
vudname 60
C:\MSC.Software\CAEfatigue_Beta\Windows\Resources\Materials\cfv_material_db.xml
fe_units 777 MPA 1 B N mm
vftgdef 777
timeout S
histrcc elset
fset3 10001 ELEM 10001
13247
vftgparm 777
stress sgvon none naver none 1E-05
dsp filter peakvaly
vftgseq 777 1 seconds 1 yes
101 102 1
vftgevent 101 flight_1
vftgevent 102 flight_2
casemap C:\A_presale\Industry_based\Digimat\CF\QW22001_Demonstrator_CAEfatigue_DigimatCoupled.hs
1 subcase 1, step 1
vftgload 402 nqstatic 0.0005
vftgload 403 nqstatic 0.0001
C:\A_presale\Industry_based\Digimat\CF\QW22001_Demonstrator_CAEfatigue_DigimatCoupled.hs
C:\A_presale\Industry_based\Digimat\CF\QW22001_Demonstrator_CAEfatigue_DigimatCoupled.hs
```



Jump to Page:	Go	Rows/Page:	50	Displayed Rows:	1 - 2 of 2	Selected Rows:	0
Item	Variant	Project	Phase	Scenarios	Model File	Created At	
Mainframe_Location2	Simulation_Mainframe	V1	/Data/Flender/Mainframe2_Simulations	Concept	[Fatigue]	mainframe_at_location_2.txt (1058 b)	13.03.2023; 22.08.46
Mainframe_Location1	Simulation_Mainframe	V1	/Data/Flender/Mainframe2_Simulations	Concept	[Fatigue]	mainframe_at_location_1.txt (1058 b)	13.03.2023; 22.02.53

New 2023.2 PROCESS FLOW TEMPLATE for Aerospace Loads Spectrum

Defined Load spectrum files, fatigue parameters, select solver results press RUN and see results.

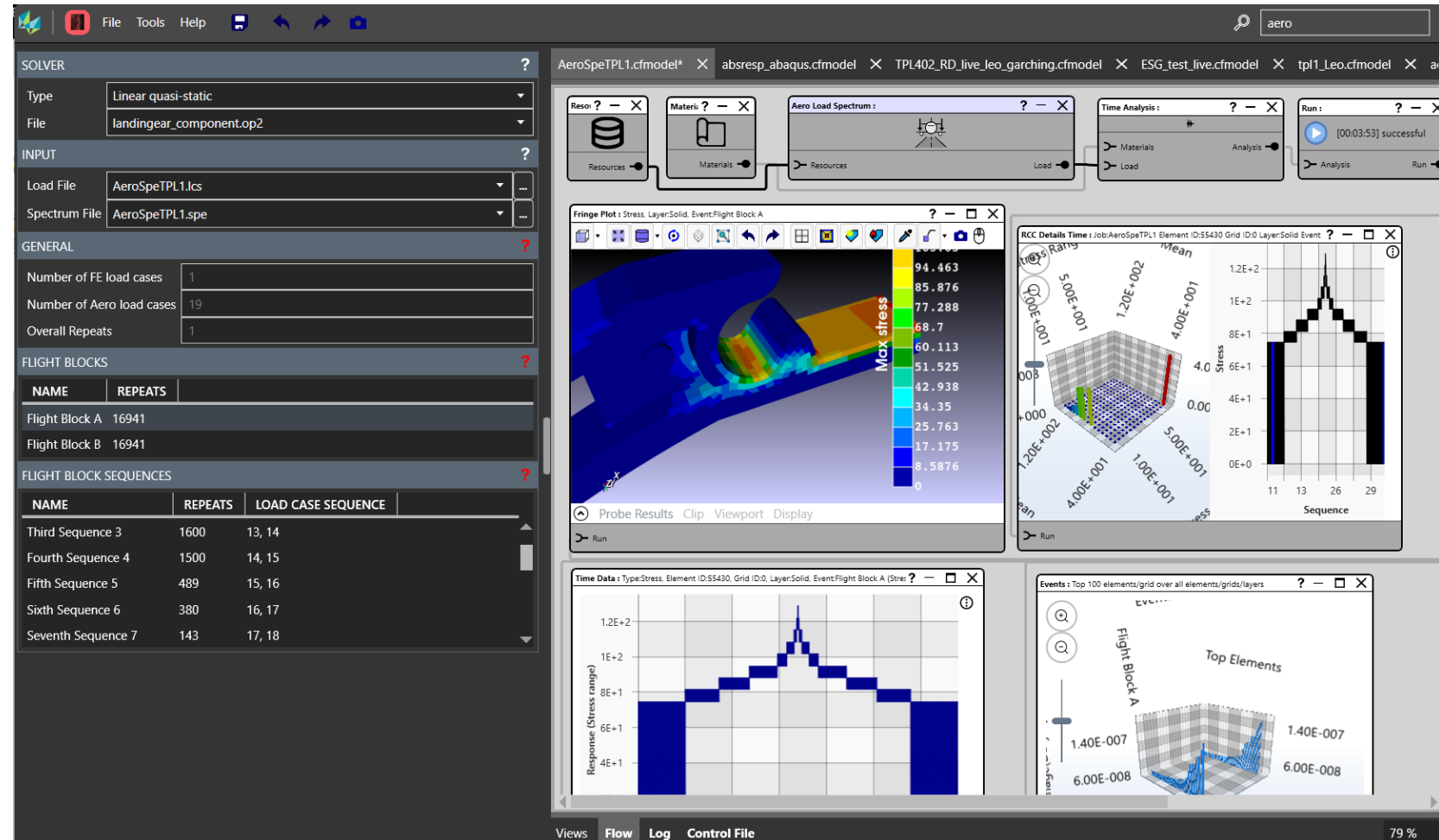
Flight Mission Spectrum Files

• Load cases file (.lcs)

```
1 LOAD FILE V2 # LOAD FILE V2
2 Title # This is a title
3 Example # Description
4 MPa # units
5 19 # number of aero load cases (new ids)
6 1 # number of FE cases
7 1.0 # divisors
8 # in case of mecaer not wanting to make linear combinations
9 # use a identity matrix for SCALARS
10 11, Zero1, 0 # internal id, name, SCALARS
11 12, 0.1A, 1.1
12 13, 0.2B, 1.2
13 14, 0.3C, 1.3
14 15, 0.4D, 1.4
```

• Load sequence file (.spe)

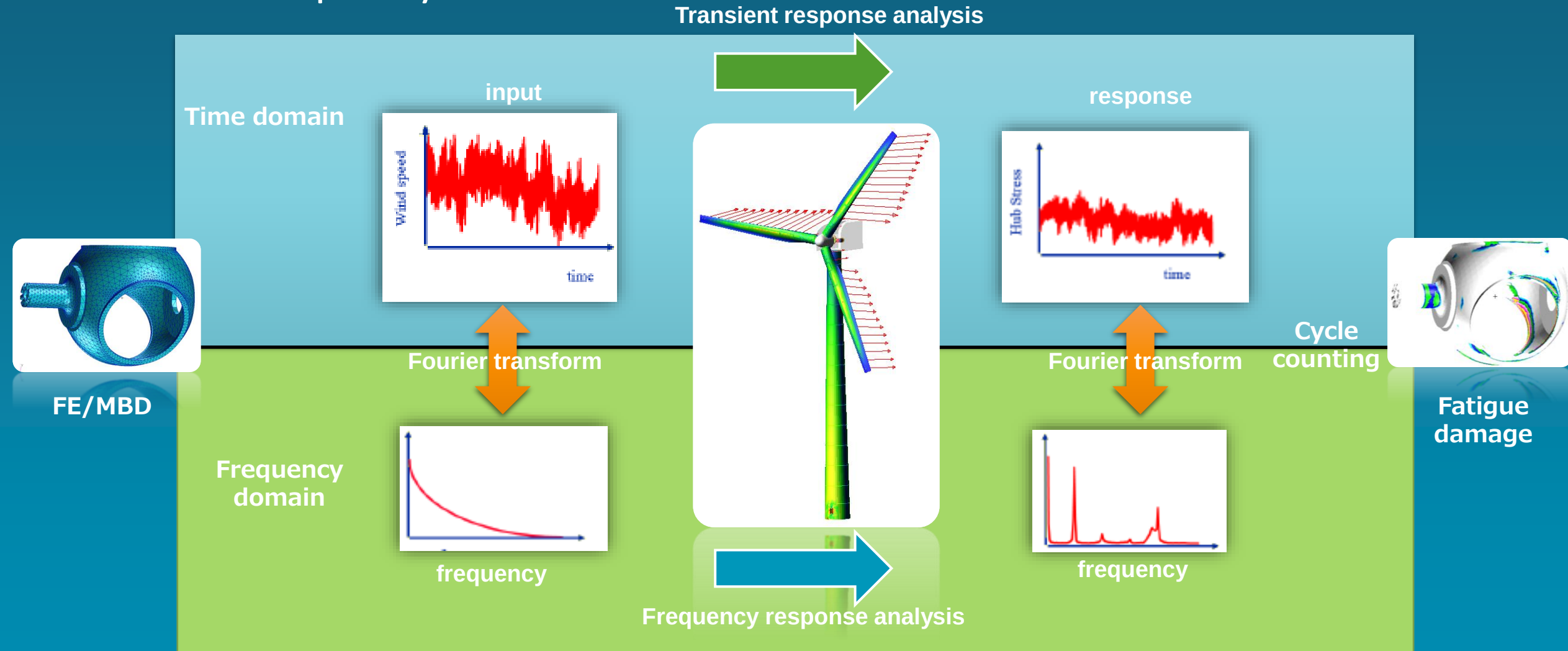
```
1 Spectrum File V3
2 Title # title
3 Description # Description
4 1 # repeats of event ?
5 16941 # total number of flights
6 2 # Number of flight blocks
7 0 # Stress increment
8 #
9 # flight blocks
10 #
11 Flight Block A, 1, 19, 16941 # name, uni
12 First Sequence 1, 2500, 11,12 #
13 Second Sequence 2, 1789, 12,13
14 Third Sequence 3, 1600, 13,14
15 Fourth Sequence 4, 1500, 14,15
16 Fifth Sequence 5, 489, 15,16
17 Sixth Sequence 6, 380, 16,17
18 Seventh Sequence 7, 143, 17,18
```



Results support:
New histogram of most damaging pairs

Fatigue analysis in dynamically responsive structures

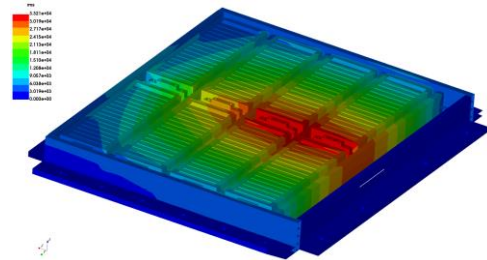
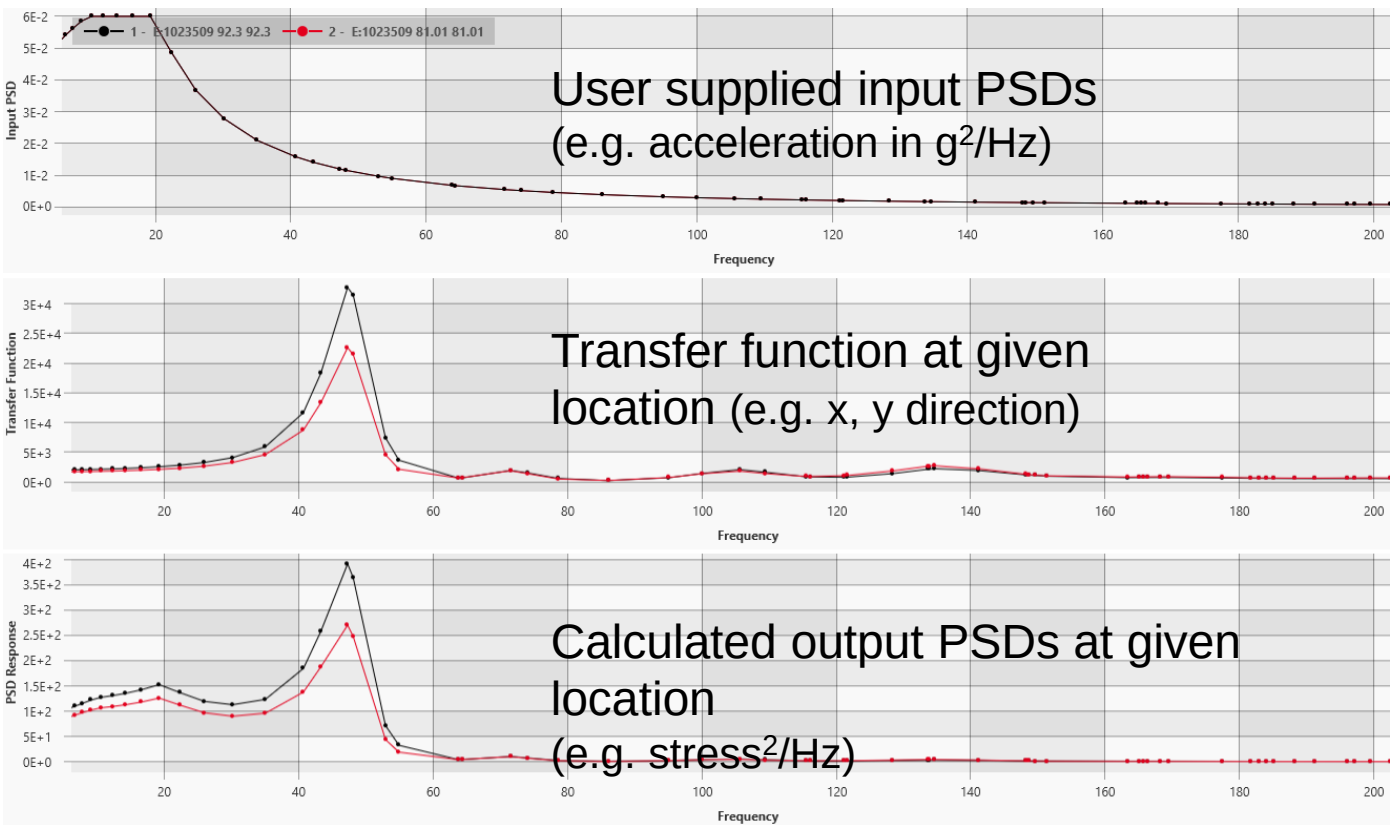
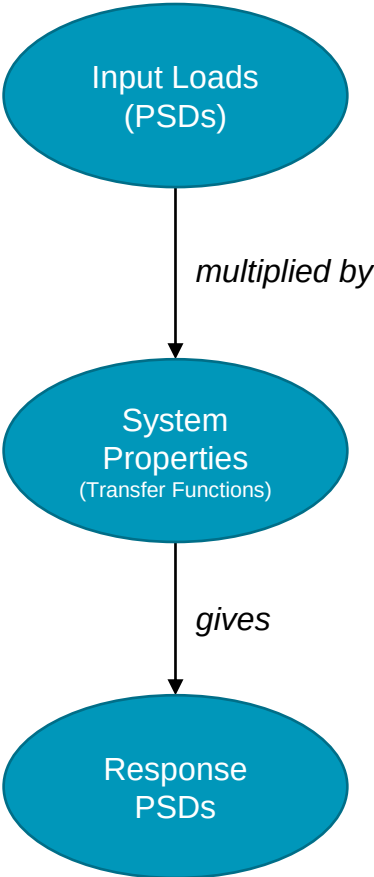
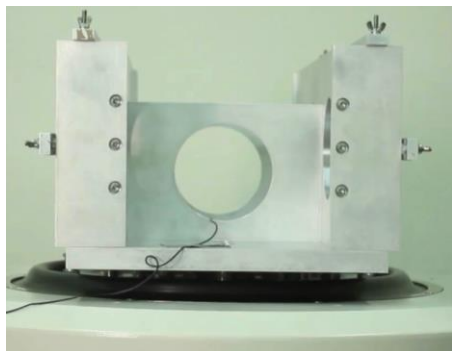
Time vs frequency domain



Vibration fatigue fundamentals: single input base shake

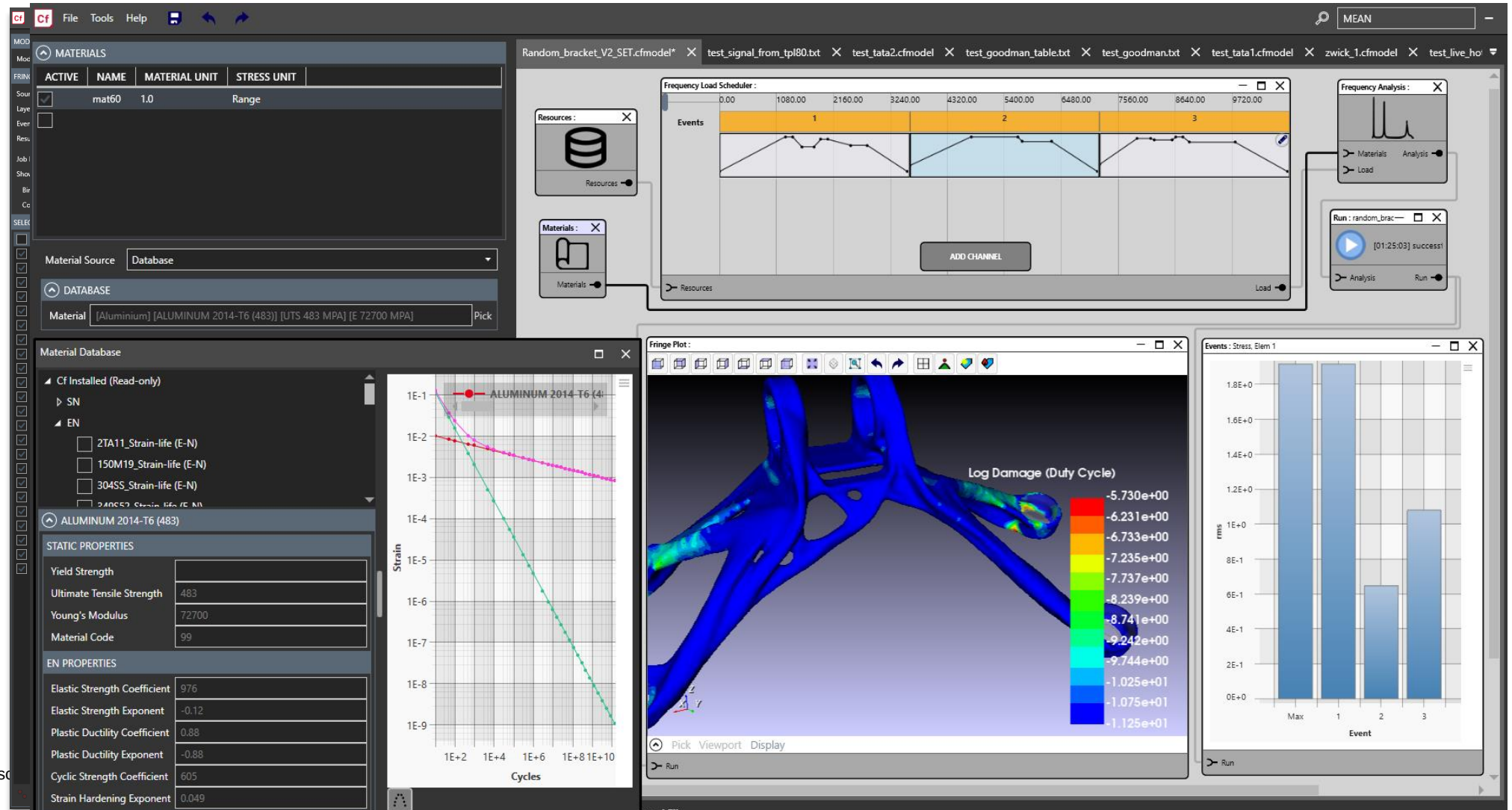
Visual random and vibration workflows

Single Input Base



PROCESS FLOW - BASE SHAKE USING RANDOM TEMPLATE

Define fatigue parameters, define loading, select solver results press RUN and see results.



Visual Random response with CAEfatigue Process flow

User guide battery support structure example

Multiple loading conditions including

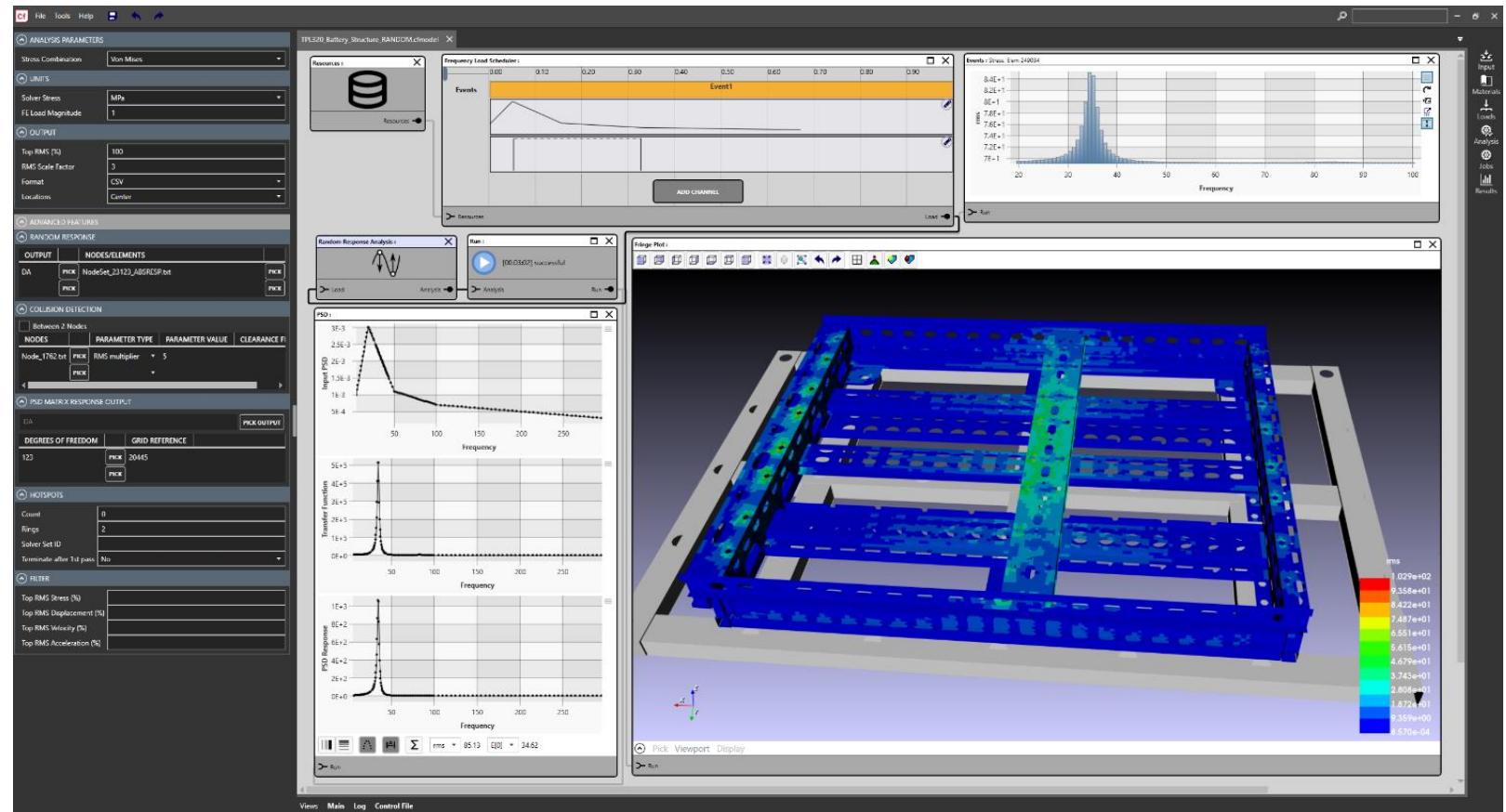
- Random only
- Sine sweep only
- Combined Random and Sine Sweep

Results support

- Both Stress and DVA analysis
- Collision detection (rattle and wear)
- Load cascading

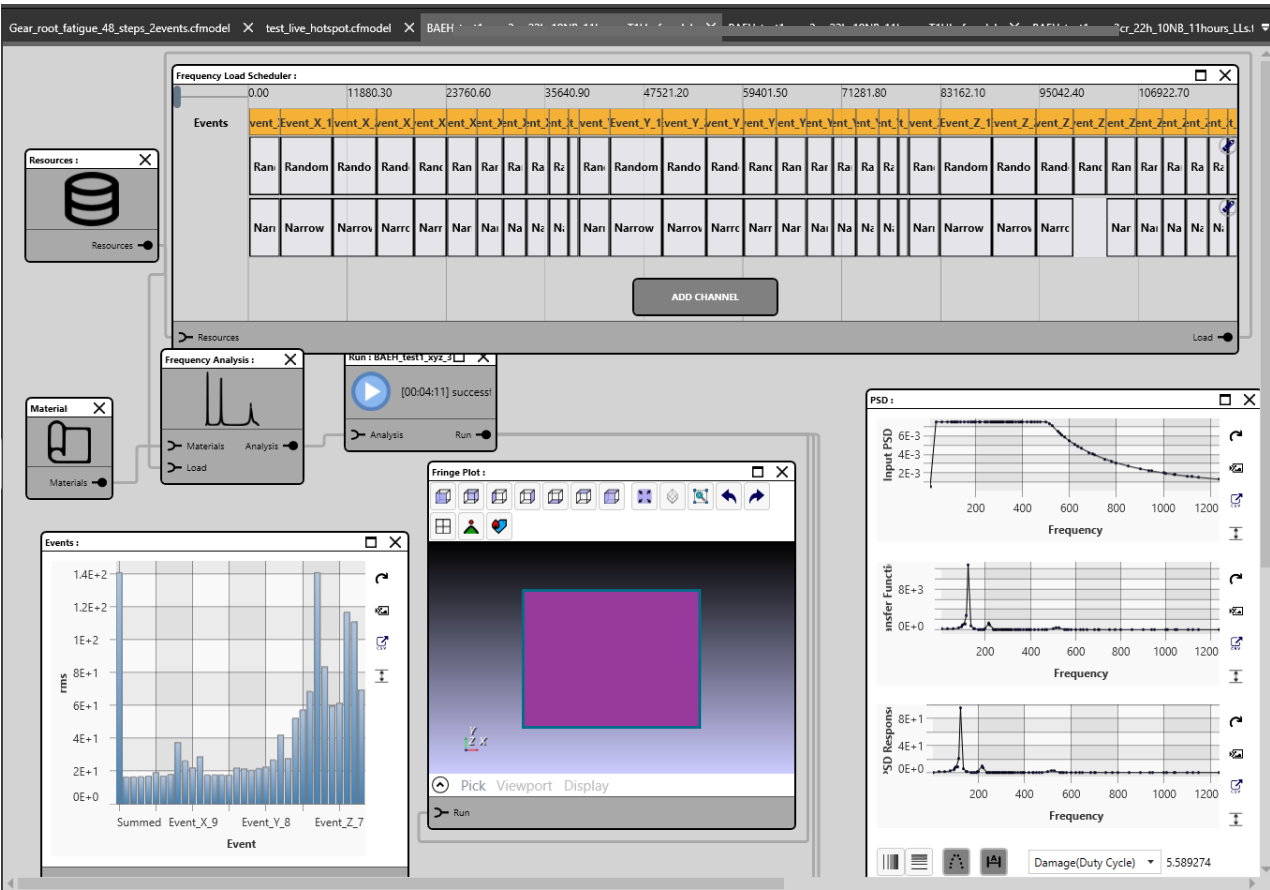
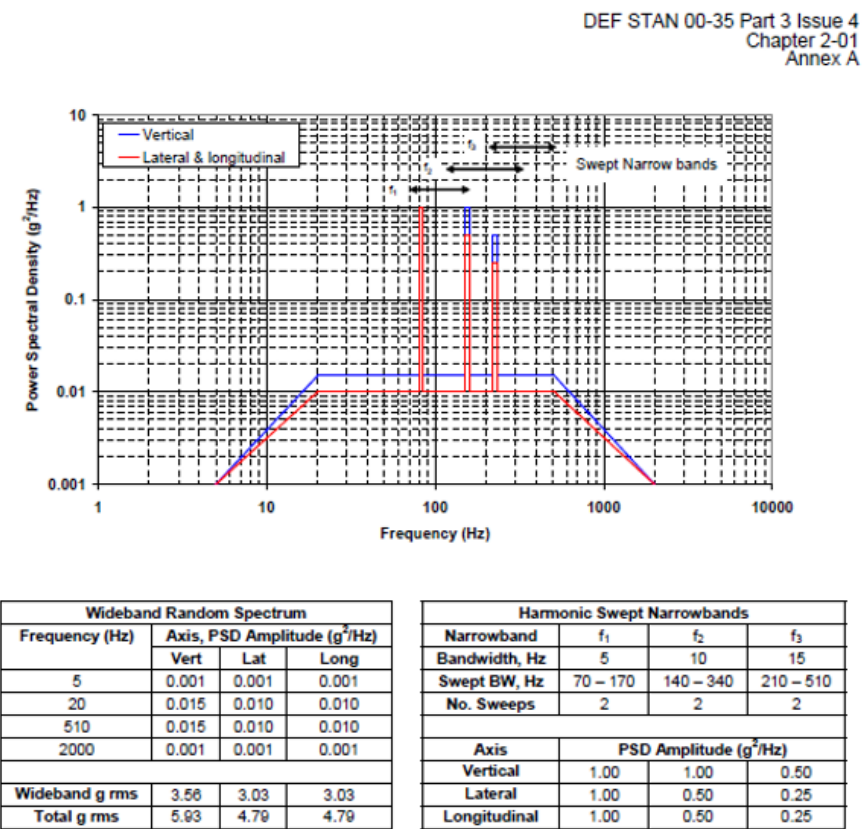
Model Filtering

- Hotspot detection



Applications: Part design and validation fatigue simulation – e.g. MIL-STD-810

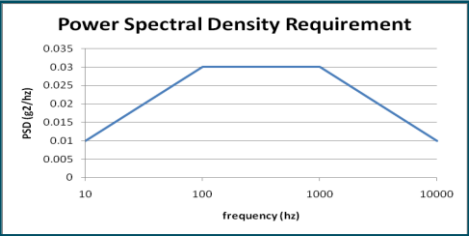
Narrow Band or sine sweep on Random loading (e.g. random gunfire or rotors on vibration)



Advanced **Mixed Loading** required for aerospace and automotive applications

Random Vibration

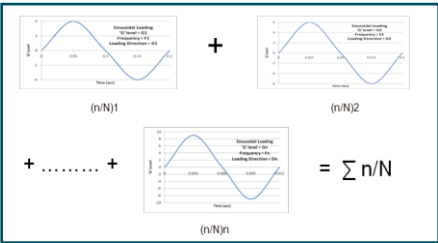
- * Each direction of loading has an independent:
 - PSD level
 - Clipping level
 - Length of testing



Consecutive Sines ('windmilling condition')

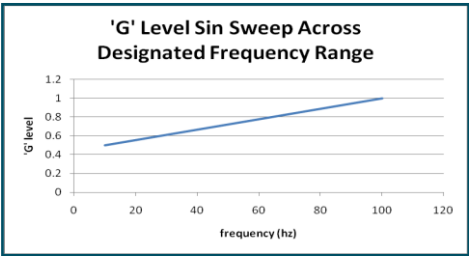
(with or without a consecutive random vibration loading condition)

- * Each direction of loading has sinusoids with independent:
 - Magnitude
 - Frequency
 - Length of time



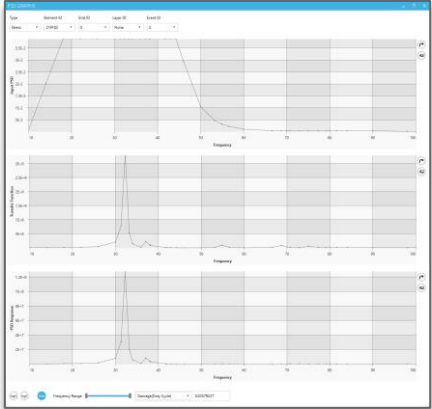
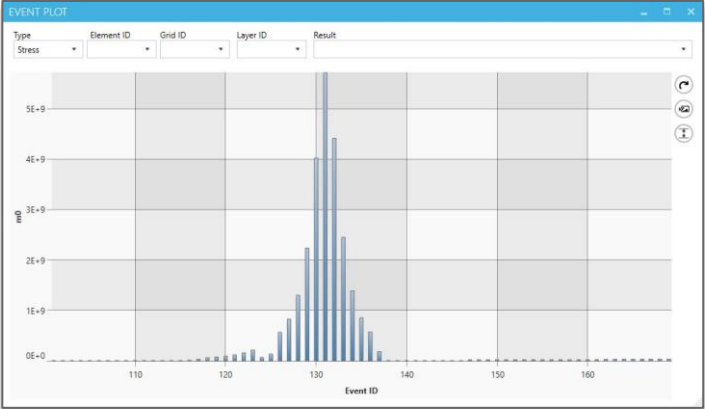
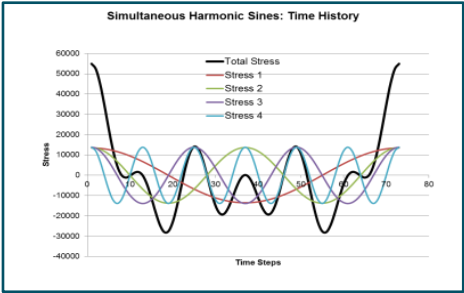
Sine Sweep

- * Each direction of loading has an independent:
 - G loading
 - Sweep rate
 - Number of passes
 - Sweep range



Simultaneous Sines (with or without random background)

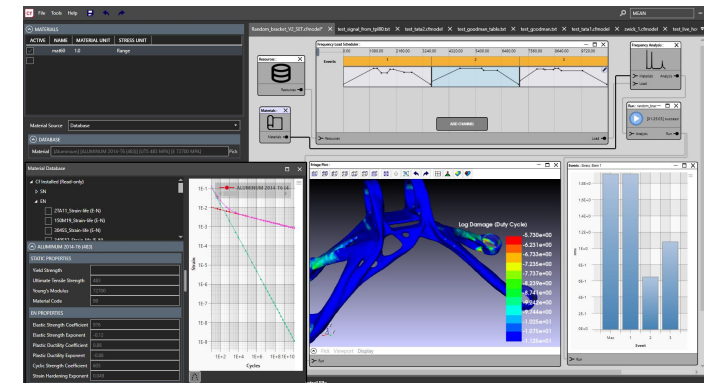
- * Each direction of loading has an independent:
 - Length of Test
- * Each direction of loading has sinusoids with independent:
 - Magnitude
 - Frequency
- * If required, each direction of random vibration loading has independent:
 - PSD level, Clipping level, Length of Testing



...From challenges to solutions

Robust Design & CAEfatigue value *what makes it unique*

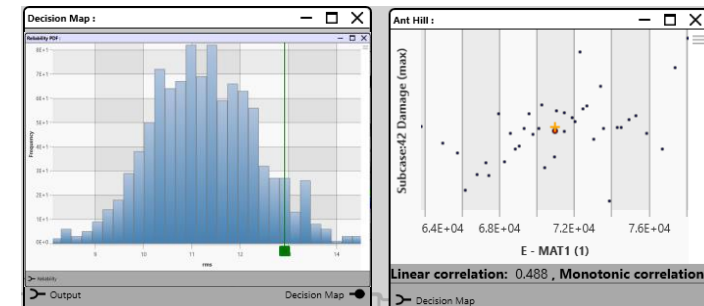
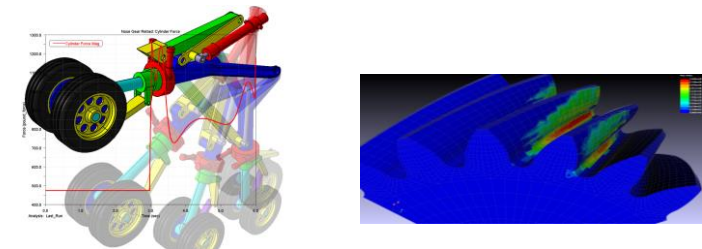
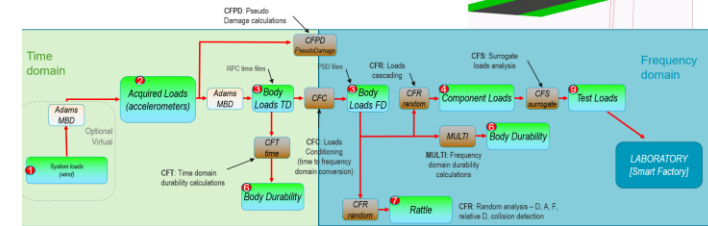
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The Loads Management Process

CAEfatigue supports the load management process throughout all the stages of the design, engineering and testing.

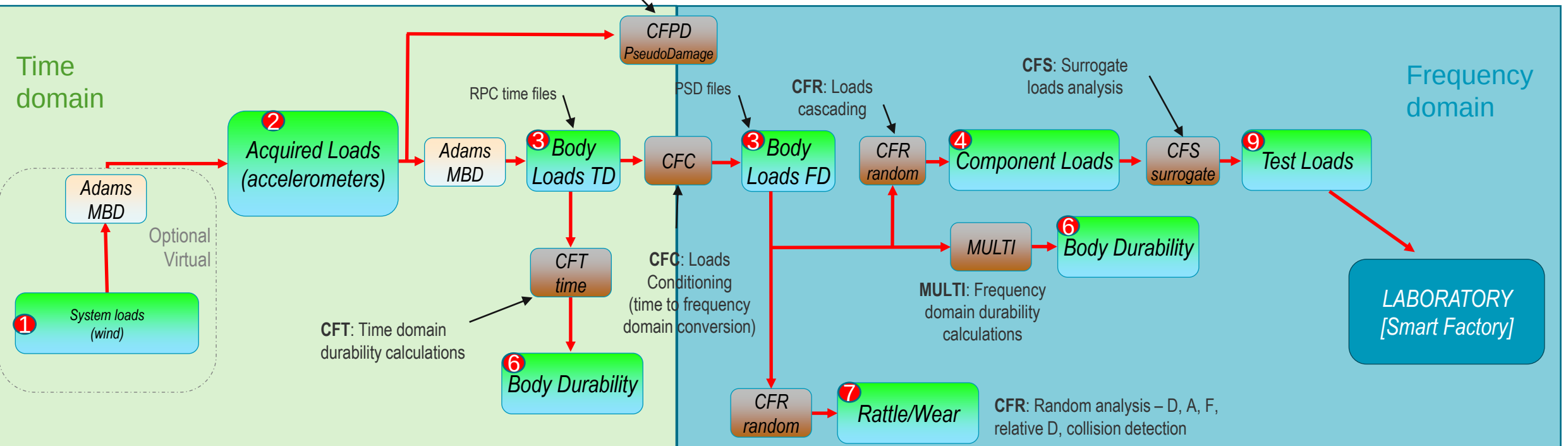
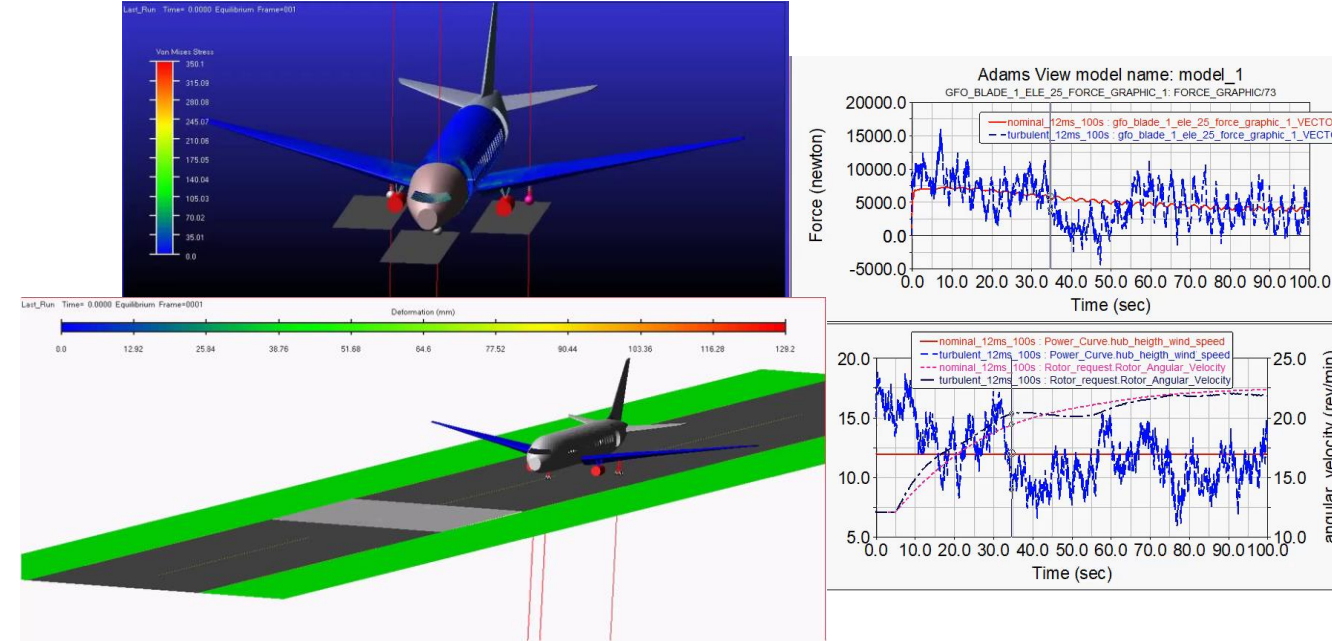
From assessing event and configuration severity directly on the acquired signals (Pseudo Damage analysis) through Random and fatigue analysis and onto test optimization via loads enveloping and surrogate loads identification.



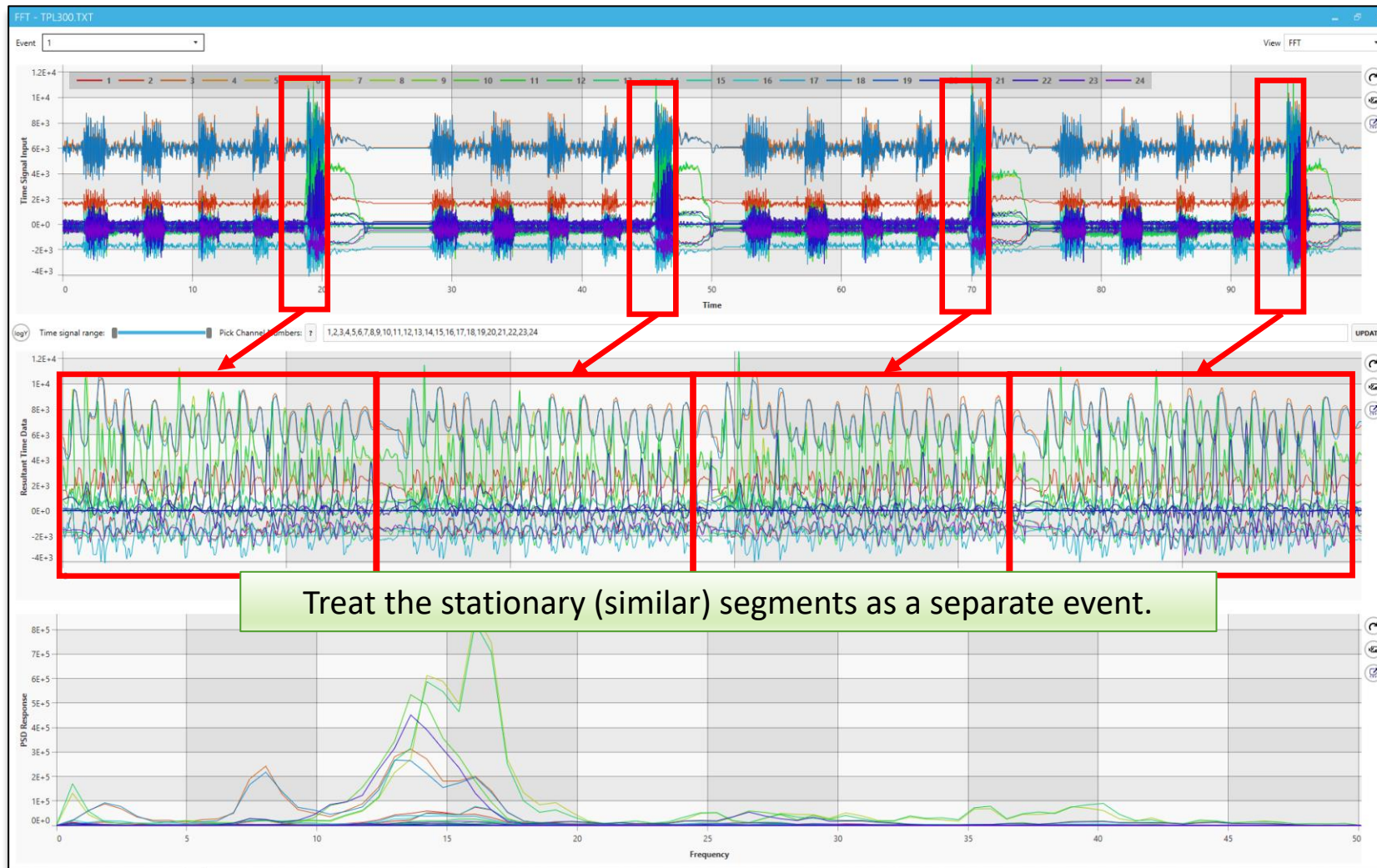
The Loads Management Process

CAEfatigue supports the load management process throughout all the stages of the design, engineering and testing.

From assessing event and configuration severity directly on the acquired signals (Pseudo Damage analysis) though Random and fatigue analysis and onto test optimization via loads enveloping and surrogate loads identification.



CAEfatigue for Loads Conditioning

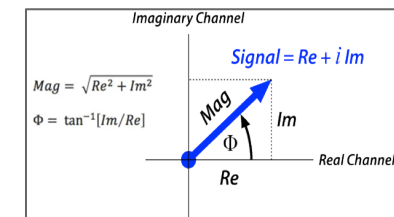


AutoD: When used with sub events (subev) this option enables statistically similar sections in the original time data (based on the rms) to be collated and cleaned up into one consistent data sample.

AutoT: Is a powerful algorithm which makes the choice of FFT buffer length (T) much more user friendly, and easy to do for all events simultaneously.

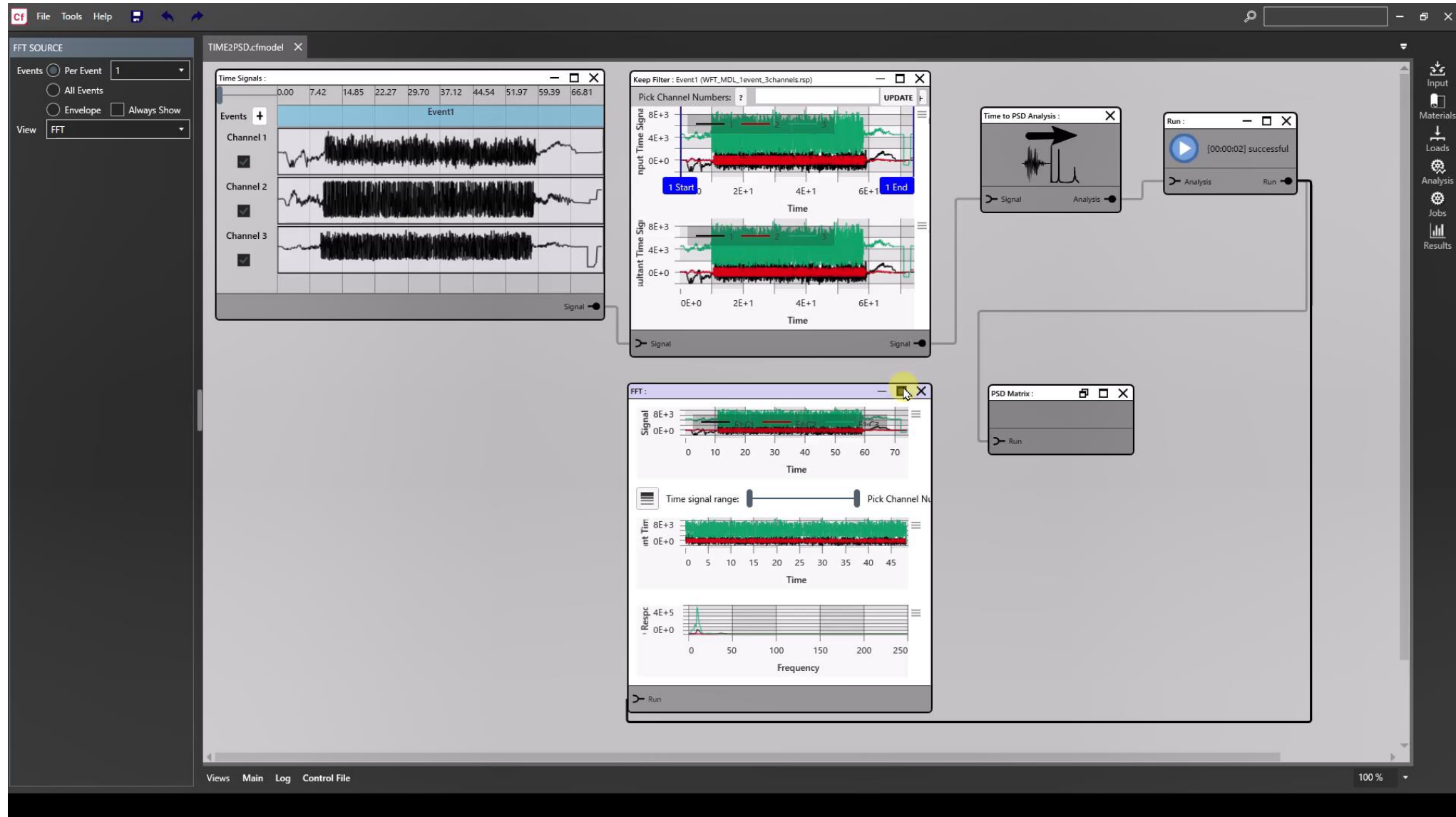
Direct PSD 1-1	Cross PSD 1-2	Cross PSD 1-3	Cross PSD 1-4
Cross PSD 1-2	Direct PSD 2-2	Cross PSD 2-3	Cross PSD 2-4
Cross PSD 1-3	Cross PSD 2-3	Direct PSD 3-3	Cross PSD 3-4
Cross PSD 1-4	Cross PSD 2-4	Cross PSD 3-4	Direct PSD 4-4

Relationship between real / imaginary and magnitude / phase.

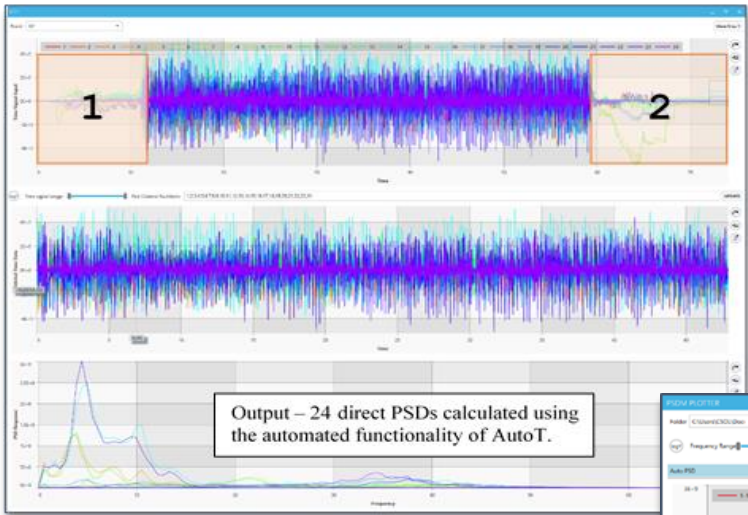


PSD Matrix (PSDM)

Time2PSD: automatic conversion from time to frequency domain

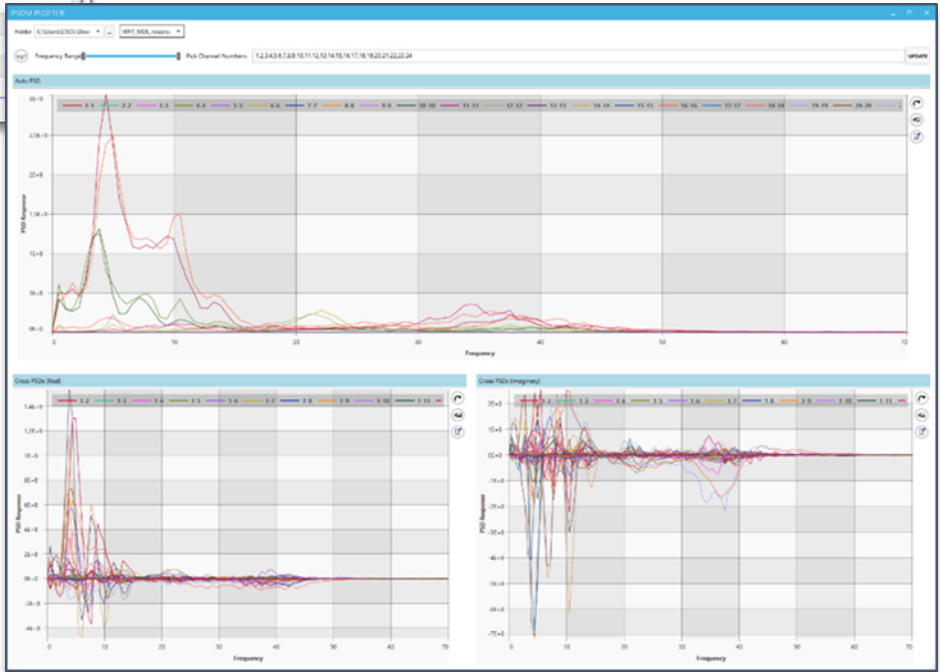
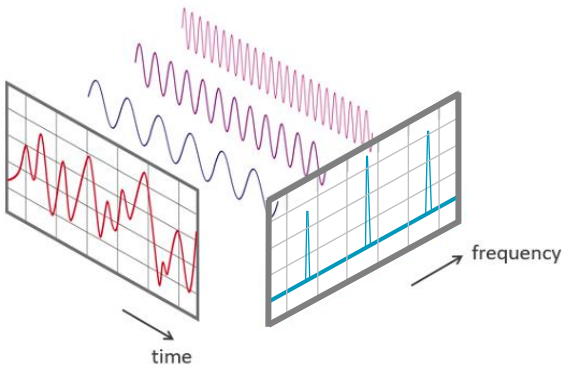


Advanced **Loads Conditioning** to convert time domain data to frequency domain



Creation of PSD Matrix

Direct PSD 1-1	Cross PSD 1-2	Cross PSD 1-3	Cross PSD 1-4
Cross PSD 1-2	Direct PSD 2-2	Cross PSD 2-3	Cross PSD 2-4
Cross PSD 1-3	Cross PSD 2-3	Direct PSD 3-3	Cross PSD 3-4
Cross PSD 1-4	Cross PSD 2-4	Cross PSD 3-4	Direct PSD 4-4



NAFEMS World Congress 2019 – NWC19-378
Paper Title: Loads Conditioning for Frequency Domain Analysis

Stuart Kerr and Neil Bishop
(CAEfatigue Limited, UK);
Sandip Datta
(Fiat Chrysler Corporation, USA).

Abstract
The accurate conversion of time-based data into an equivalent frequency domain form has become very important because of the increased use of frequency domain analysis for random response and durability calculations. Typically, for road vehicles, the loading conditions are measured at the wheel center and then cascaded into body loads using multi body dynamics systems (a partially virtual model). Or, in a more sophisticated system, the road profiles could be combined with a wheel/tire model to transfer the loads through the wheel system and onto the body of the vehicle (a fully virtual model). In both simulations, the cascaded loads on the body are in the time domain and usually in the form of multiple channel time signals. The correlation between the multiple channels can be very important to the final outputs for random response and fatigue.

If a frequency domain analysis is required, this time data must be converted into the frequency domain. This involves well understood “Fourier” conversion techniques in the form of Fourier Series or Fourier Transforms. However, a typical analyst faces 3 significant barriers when performing this conversion process.

Firstly, the frequency domain approach itself requires adherence to limiting assumptions. The data being processed must be,

- Stationary
- Gaussian
- Random

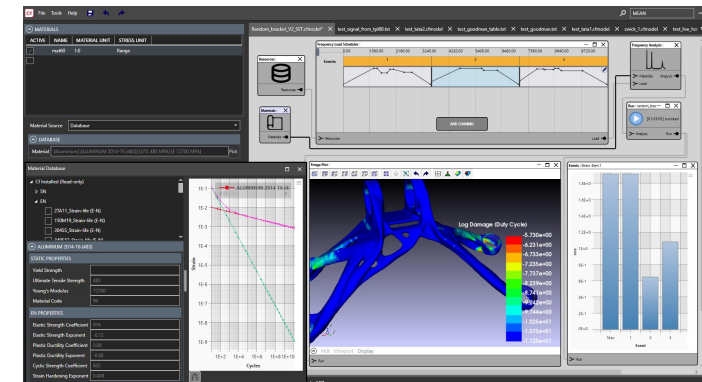
Typical users struggle to quantify these assumptions.

Secondly, the legacy use of Fourier Transforms (such as the Fast Fourier Transform – FFT) is difficult for a typical User because of several variables which must be set (such as FFT window shape, FFT window length, etc.). Setting these (and other variables) requires prior experience, which is beyond a typical User. Furthermore, the variables usually have to be set one by one for each load set (event) and this can be impractical.

...From challenges to solutions

Robust Design & CAEfatigue value *what makes it unique*

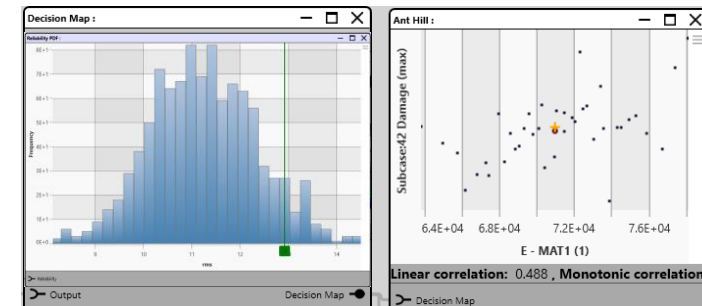
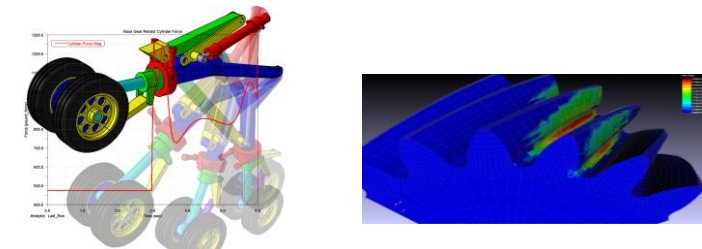
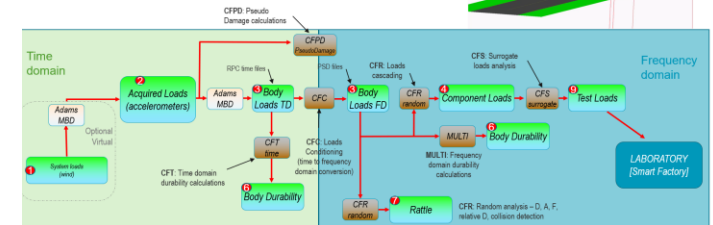
- CAEfatigue architecture
 - Fast and intuitive Process Flow templates
 - Solver neutral (Nastran, Abaqus, Ansys...)
 - batch runs and easy integration in enterprise portals (e.g. SimManager)
 - Visual Random and Vibration response tools
- Load Management Process
 - Digital Signal processing and automatic conditioning for time to PSD loading conversion
 - Using CAEfatigue for test optimization (envelopes and surrogates)
 - Virtual Strain gauge analysis
- Fatigue analysis integration in CAE processes
 - Embedded fatigue solutions, Manufacturing effects, mechanism and transmission durability, advanced materials
- Robust Design for stochastic simulation and stochastic design improvements



The Loads Management Process

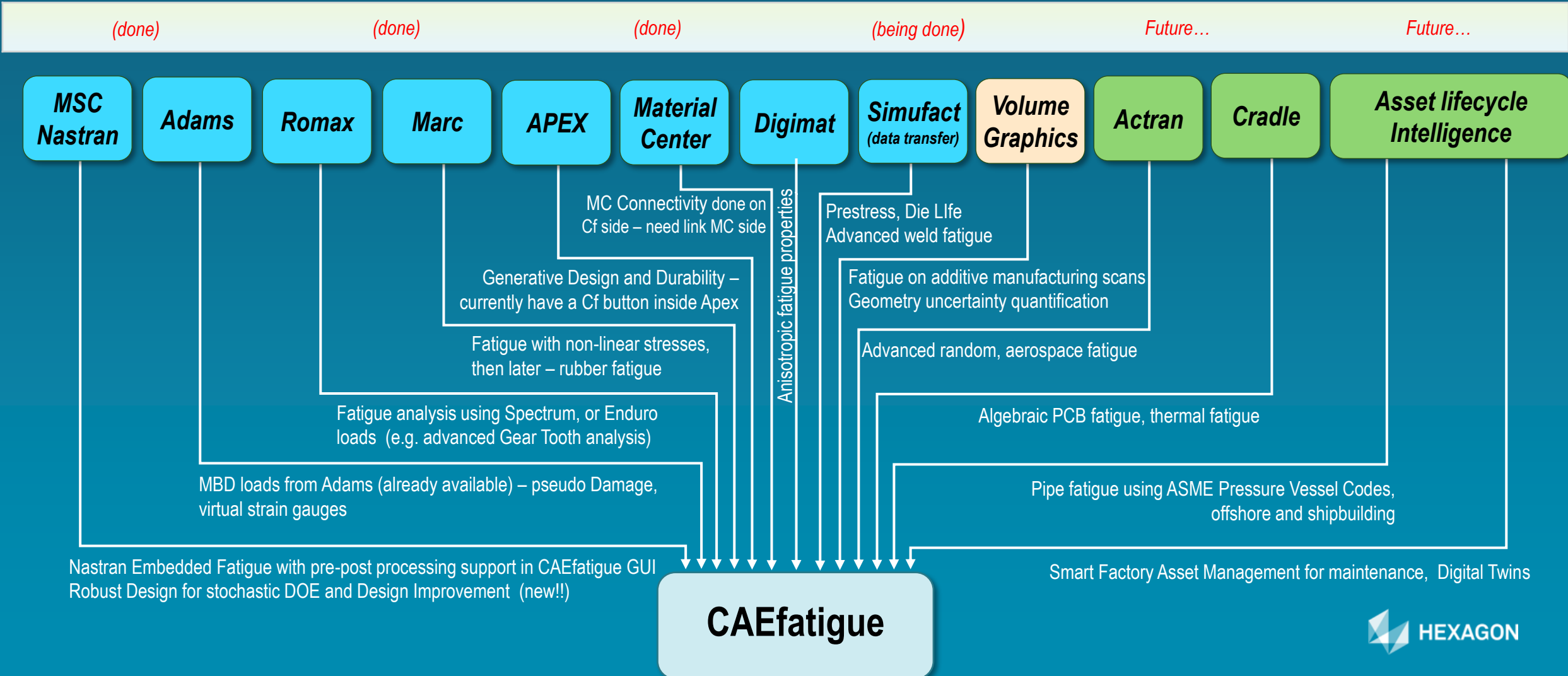
CAEfatigue supports the load management process throughout all the stages of the design, engineering and testing.

From assessing event and configuration severity directly on the acquired signals (Pseudo Damage analysis) through Random and fatigue analysis and onto test optimization via loads enveloping and surrogate loads identification.



CAEfatigue Interoperability with other Hexagon Solutions

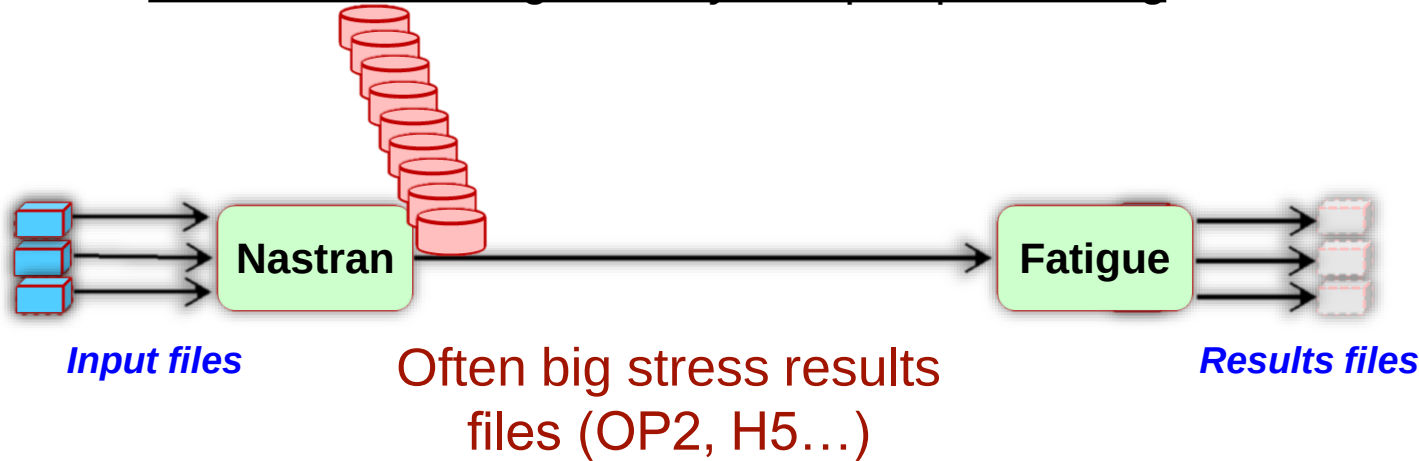
CAEfatigue and Robust Design are already integrated with many of the Hexagon solutions within the manufacturing workspace and work is ongoing to extend this interoperability as described below.



Stand alone CAEfatigue or embedded NEF(time) and NEVF (frequency)

Dual path for efficient fatigue analysis

Conventional Fatigue Analysis a postprocessing

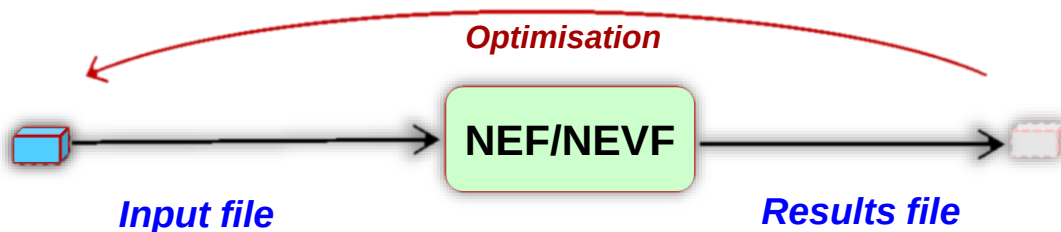


CAEfatigue Process Flow and control files

INTERACTIVE or BATCH: GUI driven
INTUITIVE: easy and repeatable workflows
WHAT IF: easy interrogate and modify
REPORTING: nested postprocessing
ASSISTED: linked training and documentation

Nastran Embedded Fatigue NEF(time) / NEVF(frequency)

Embedded Fatigue Analysis (no need for intermediate stress results)



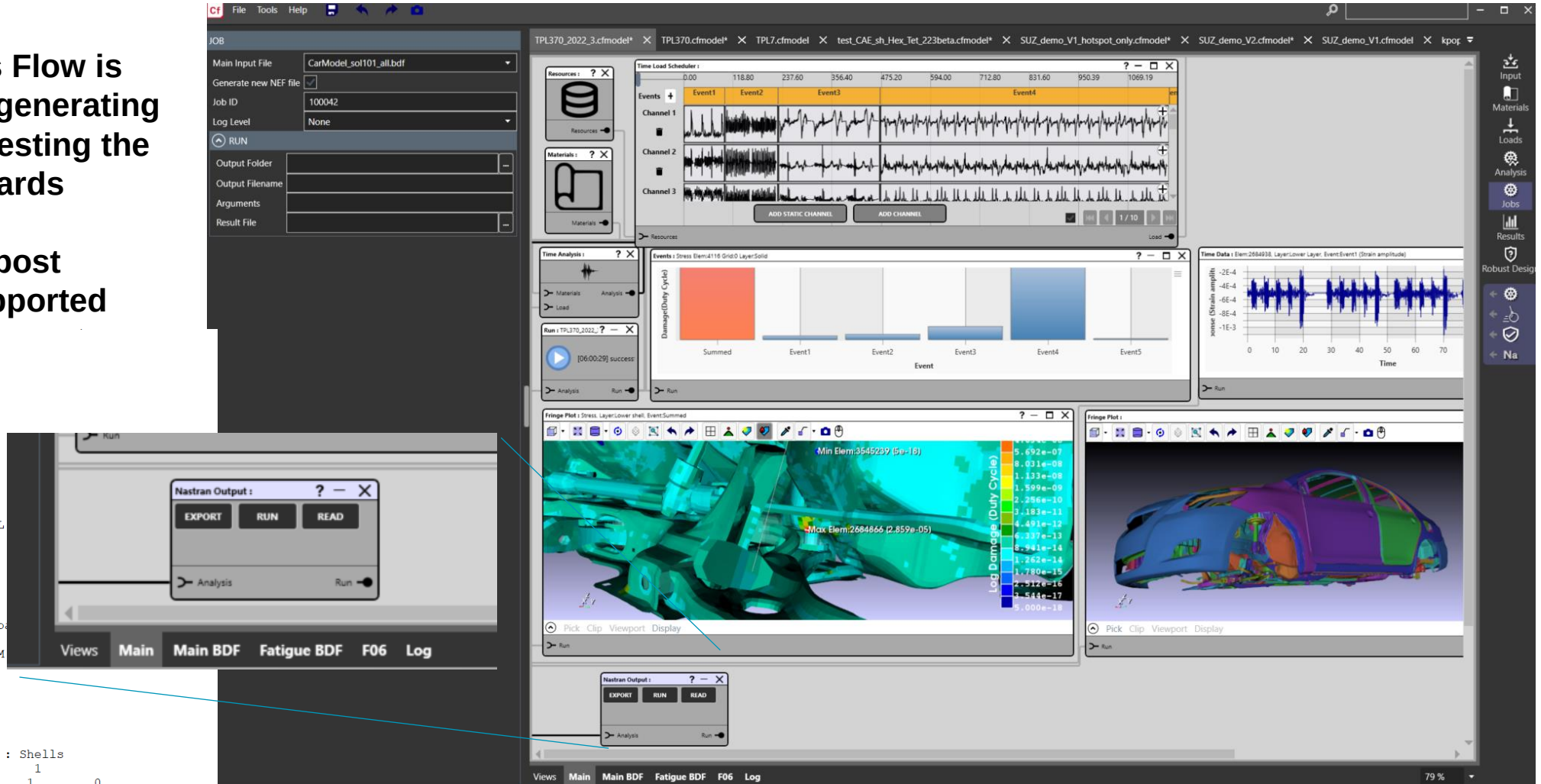
FASTER: Drastic reduction of fatigue analysis time
SMALLER: Large intermediate files eliminated
SIMPLER: combines FE and Fatigue
PORTABLE: Process embedded in the solver file
TRANSPARENT: Process is transparent (text lines)
DESIGNS Optimization: coupling fatigue analysis and FE optimization and **Robust Design**

Support for Nastran Embedded Fatigue “NEF” pre-post processing

Control File input can be directly added to an existing Nastran time domain job

- CAEfatigue Process Flow is transferred NEF by generating and automatically nesting the additional needed cards
- Interactive run and post processing also supported

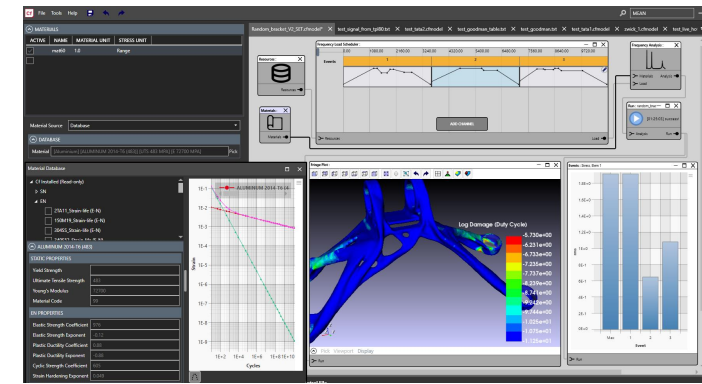
```
SOL 112
CEND
TITLE = CANTILEVER BEAM NEF MODEL
ECHO = NONE
RESVEC = NO
LOADSET = 5
FATIGUE (SORT1, FORMAT=64, BULK)=1
SUBCASE 1
  SUBTITLE=ALL
  METHOD = 1
  TSTEP = 1
  SPC = 2
  DLOAD = 6
  STRESS (PLOT, SORT2, REAL, VONMISES, BILIN)=ALL
$ Direct Text Input for this Subcase
BEGIN BULK
PARAM POST 1
PARAM GRDENT 0
PARAM PRTMAXIM YES
DTI UNITS 1
$ SN Analysis, Load Sequence: Simultaneous_Load
FTGPARM 1 SN 1. 1 0
STRESS ABSMAXPR GOODMAN ELEM
CERTINTY50.
FTGSEQ 1 0 0
UNITS 1. Repeats
TSTEP 1 108 1
EIGRL 1 10 0
$ Elements and Element Properties for region : Shells
PSHELL 1 1 .001 1 1
PFTG 1 1 NONE 1. 1. 0.
```



...From challenges to solutions

Robust Design & CAEfatigue value *what makes it unique*

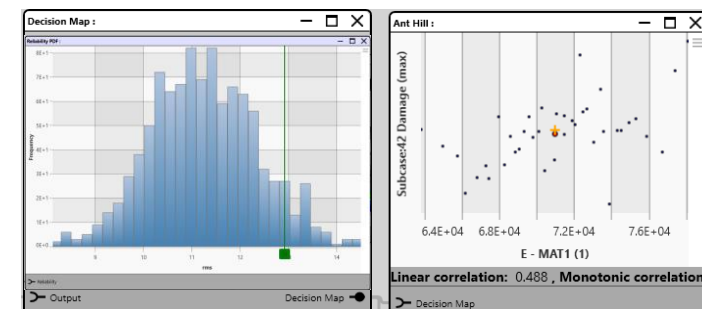
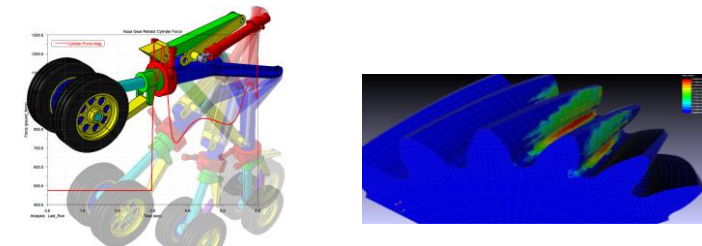
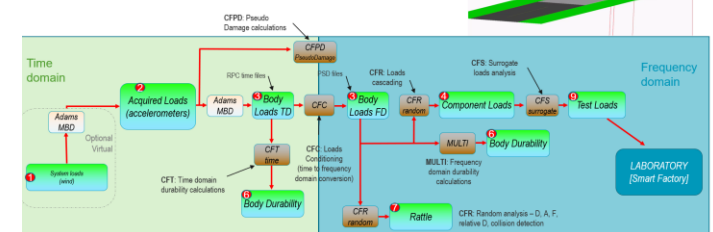
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Robust Design

Incorporating uncertainty by design

FUNCTIONALITY

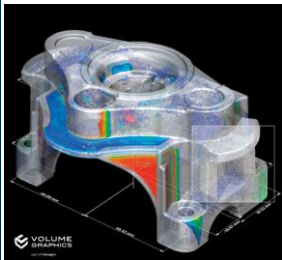


USABILITY

INNOVATION

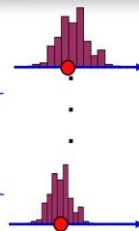
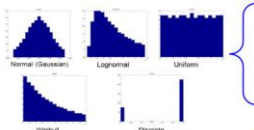
DEFINITION

- **Uncertainty** exists in everything from dimensions to material properties, loads, and boundary conditions. The incorporation of these uncertainties leads to models of unprecedented levels of realism and credibility.
- **OEMs strive to create improved and more robust designs** by identifying potential weak points, susceptible to variability
- Defining **appropriate quality controls** to be implemented in manufacturing plants



Input randomization

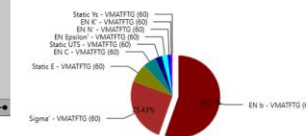
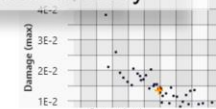
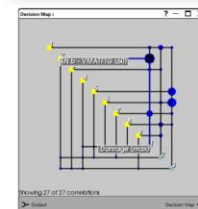
Inputs
(model properties pdfs)



RECIPE

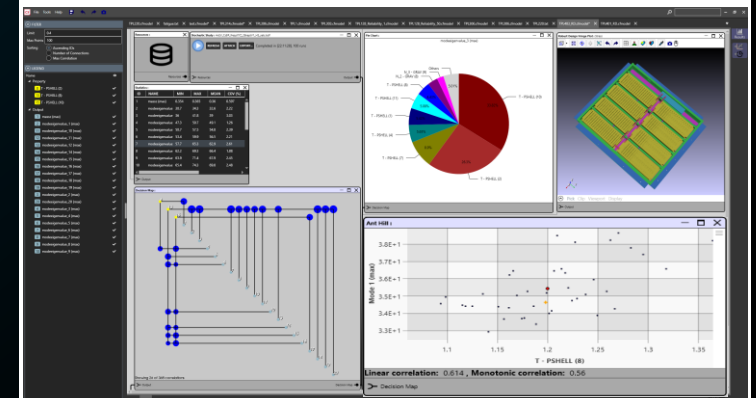
- **MSC Nastran + CAEfatigue technologies**
 - Monte Carlo DoE with intuitive randomization
 - Decision Maps and pie charts describing linear or non-linear correlation between inputs
 - Ant hill plots for determination of probability of exceedance
- **Industrial CT and optical scanning** for Non-Destructive uncertainty measurement
- **Stochastic Design improvement:** Reach design targets through robust improvement cycles

Stochastic Function Study



PURPOSE & BENEFITS

- **Simulation Health checks:** assess model numerical quality
- **Design and engineering beyond margin of safety and certification**
- **Reduce warranty and product liability costs** by identifying, correcting unwanted behaviors
- **Improving Correlation** with physical test or operational events/conditions



Stochastic Simulation Workflow: randomization and function study

- Nominal FE: a given FE model corresponding to an assumed nominal reference point
- Input Randomization: specify the tolerances and variable scatter distributions on all FE inputs subjected to variability, including customer usage, manufacturing, load factors, material and geometrical
- Stochastic function study: a Monte Carlo simulation (MCS) where the model is executed multiple times (typically 100), randomly changing each of the input variables within the assigned limits

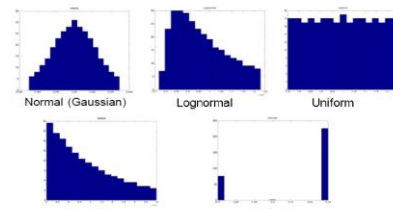
Nominal FE + Fatigue input/output model

VARIABLES	
Name	
▶ Material	
▶ Property	
▶ Connector	
▶ Temperature	
▶ Load	
▶ Output	
▶ SOL 101	
▶ Subcase 3	
▶ Subcase 4	
▶ Fatigue 100042	
▶ Global	

Name	
▶ Materials	
1 Static E - VMATFTG (60)	✓
2 EN b - VMATFTG (60)	✓
3 EN C - VMATFTG (60)	✓
4 EN Epsilon' - VMATFTG (60)	✓
5 EN K' - VMATFTG (60)	✓
6 EN N' - VMATFTG (60)	✓
7 EN Sigma' - VMATFTG (60)	✓
8 Static UTS - VMATFTG (60)	✓
9 Static Ys - VMATFTG (60)	✓
▶ Output	
1 Damage (max)	✓
3 Max Stress (max)	✓
2 RMS (max)	✓

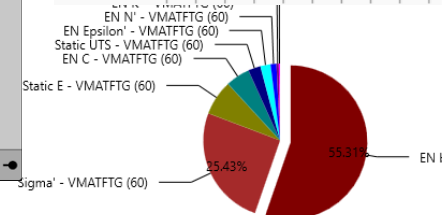
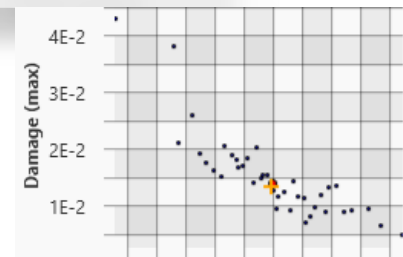
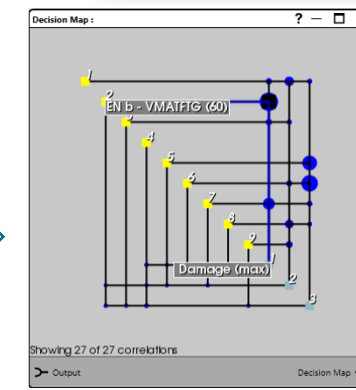
Input randomization

PSHELL (121)	✓	
PSHELL (122)	✓	
PSHELL (123)	✓	
Name	Value	Randomize
T	1.1	☒
Distribution	Gauss Normal	
Cutoff (#std.dev)	3	
Coef. of Variation (%)	5	



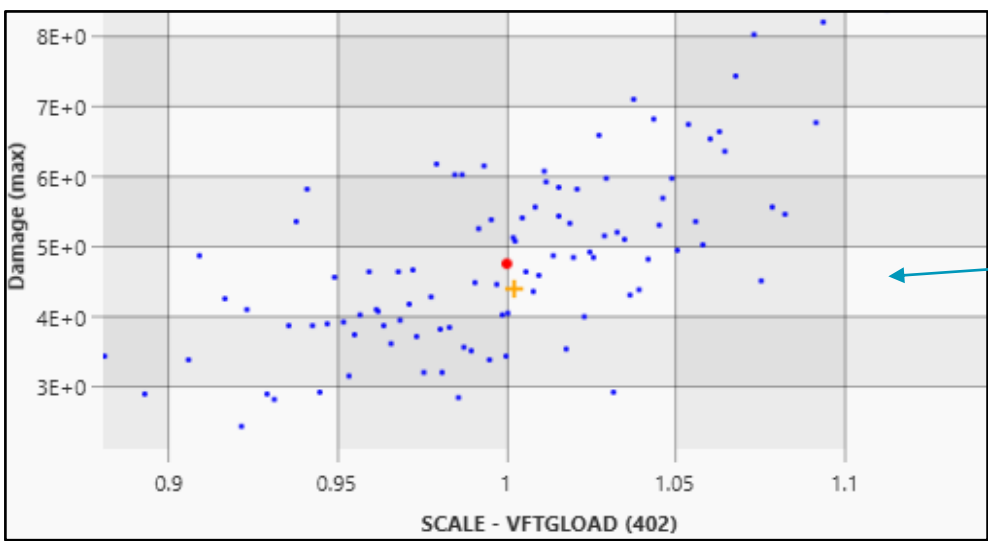
(model properties pdfs)

Function Study

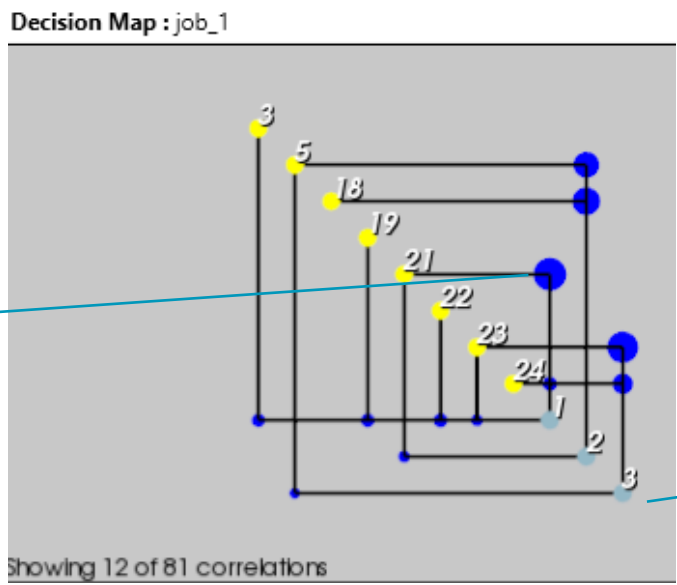


Stochastic Simulation Workflow: building actionable knowledge

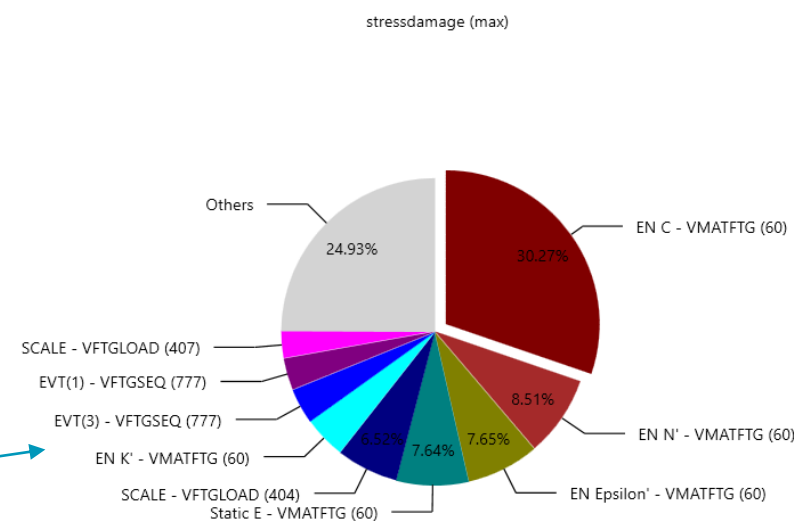
Accessible Data analysis tools enabling informed decision making about model modifications and enforcing or recommending limits for specific maneuvers and weight



Ant hill plots : distribution of input-output pairs, relationships, and correlation shape (thresholds, outliers, probability of exceedance)



Decision maps: visual input-output correlation relationship influencing the functioning of a product



Pie charts: based on correlations ranking, reflects the relative weight and influence of any parameter variation on key outputs

Robust Design to Alter Nastran Parameters and Rerun Jobs

CAEfatigue has **significantly enhanced** the feature ROBUST DESIGN, which allows you to run rerun Nastran many times while altering the input parameters by a User specified amount. Outputs show you the sensitivity of the output variable to the changing input parameters.

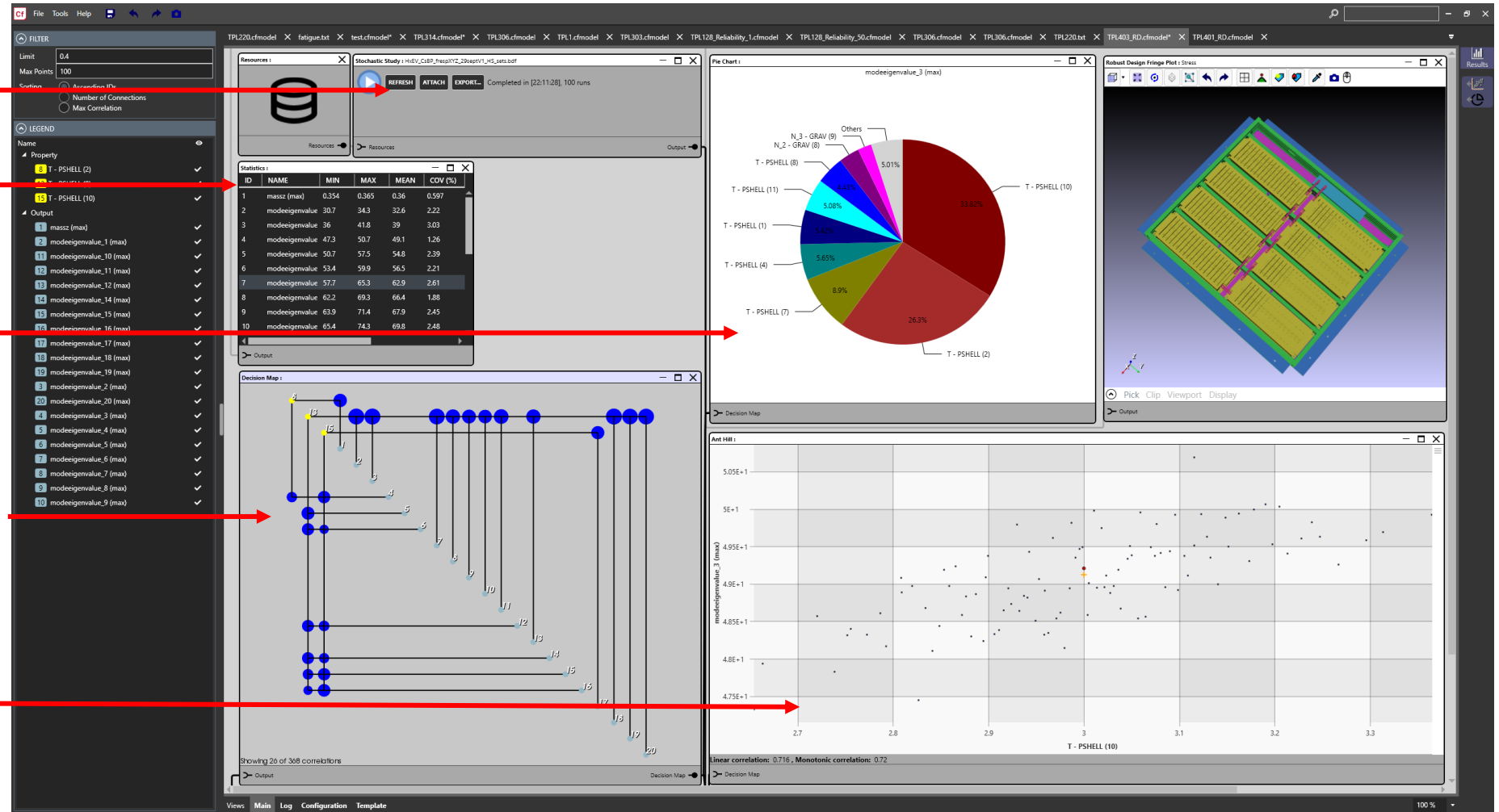
The **Stochastic Study** permits the selection of the inputs and output and controls the job operation.

The **Statistics** plot shows summary data for the Output variables

Pie charts: relative influence of the input variables on any observable output

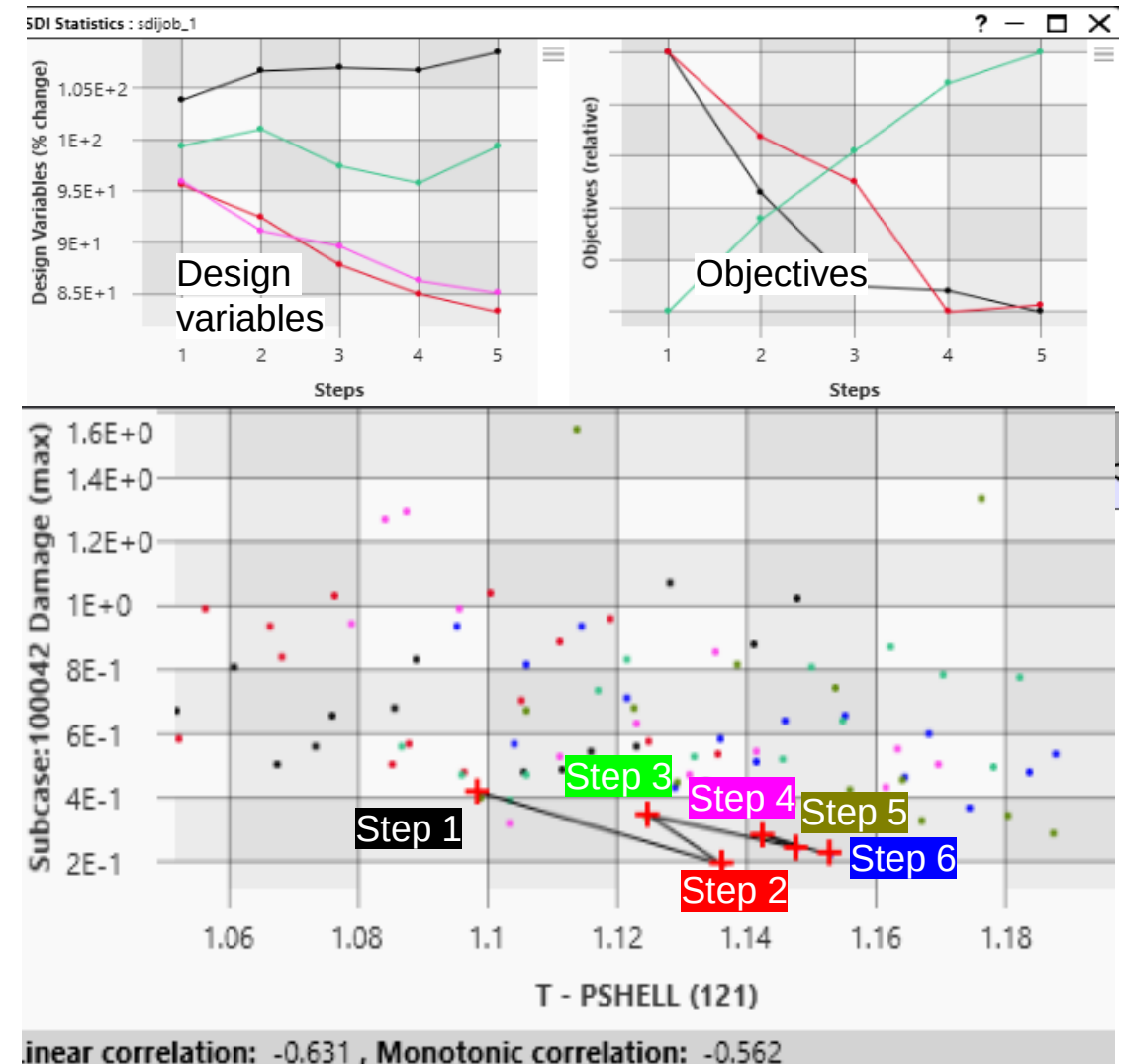
The **Decision Map** shows the input to output correlation results and how different variables influence the functioning of a product

The **ANT HILL** plot is visual distribution of input output pairs, relationships and correlation shape



Stochastic Simulation Workflow: Stochastic Design Improvement

- As a follow-on to a stochastic function study, Stochastic design improvement cycles (SDI) support gradual and robust design changes
- A gradual variation of selected design variables pushes the design towards one or more targets, achieving improved performance over the nominal analysis
 - A subset of Inputs become *Design Variables*
 - A subset of Outputs become *Objectives*
- Each stochastic simulation represents a single step. The SDI process terminates when the target *distance* is within a stated *tolerance*



Application example Robust Design to automotive durability assessment



A Stochastic Approach to Designing Robust Automotive Structures Considering Variability

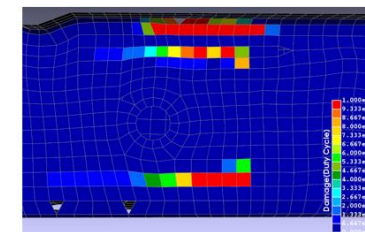
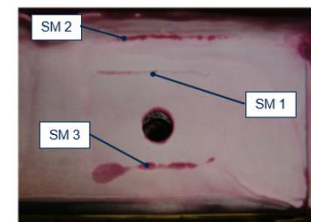


Dr. P. Roemelt,
Ford Motor Company
Dr. M. Veltri, Mark Robinson, Patrik Adolfsson
Hexagon Manufacturing Intelligence
Presenter: Jeff Robertson
Hexagon Manufacturing Intelligence



Comparison with physical test

- the fatigue analysis using nominal parameters shows good alignment with the critical areas identified in the physical test

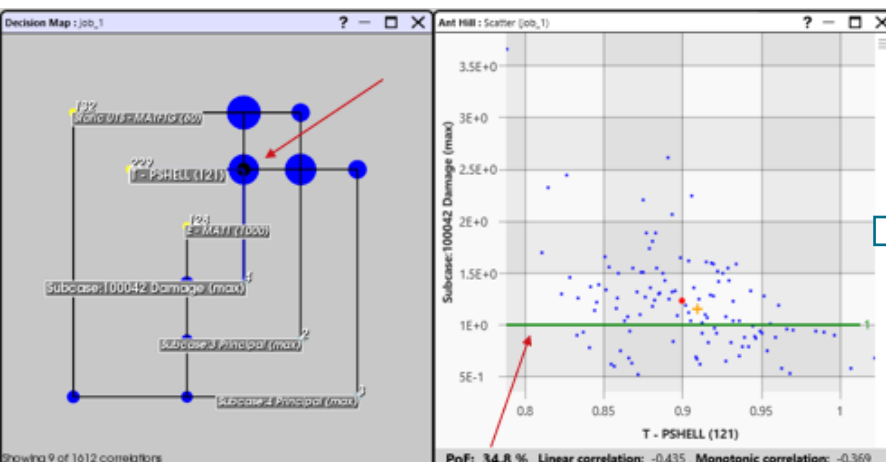


- An ensuing stochastic function study automatically incorporates uncertainties with the aim of building knowledge about model robustness

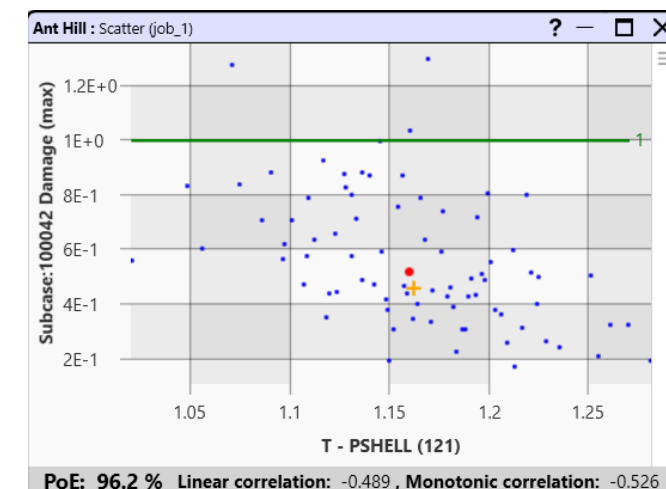
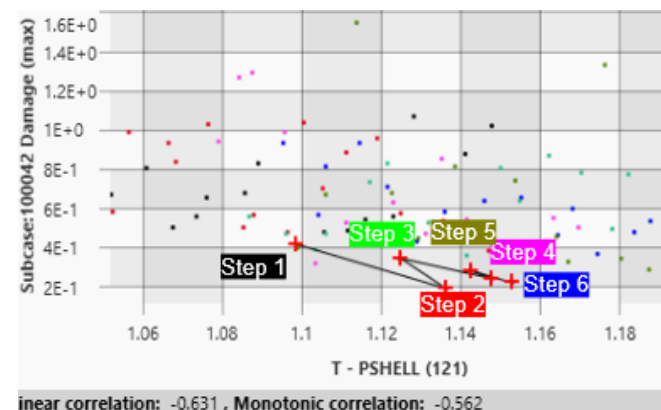


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From stochastic Design function study



To stochastic Design Improvement cycles





Incorporating uncertainty in durability assessments



Dana improves off-road durability with robust design and stochastic analysis

Dana Incorporated: Federico Bavarese (Driveline Lead Engineer), Tommaso Pirovini (Driveline Engineer), Lorenzo Bellorini (Simulation Engineer), Claudio Maffei (Virtual Verification & Integration Engineer)
Hexagon's Manufacturing Intelligence division: Marco Veltri (Business Developer), Patrik Adolfsson (Product Designer, R&D), Mark Robinson (Technical Consultant)

Challenge

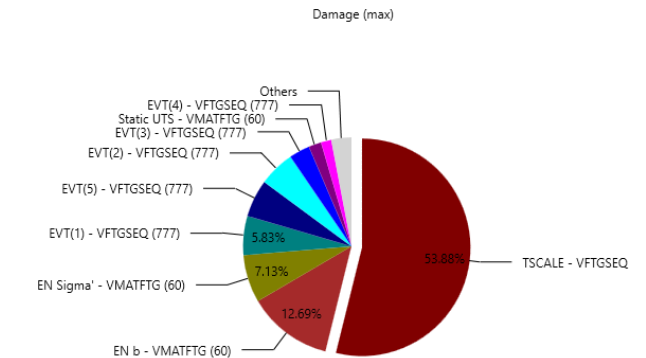
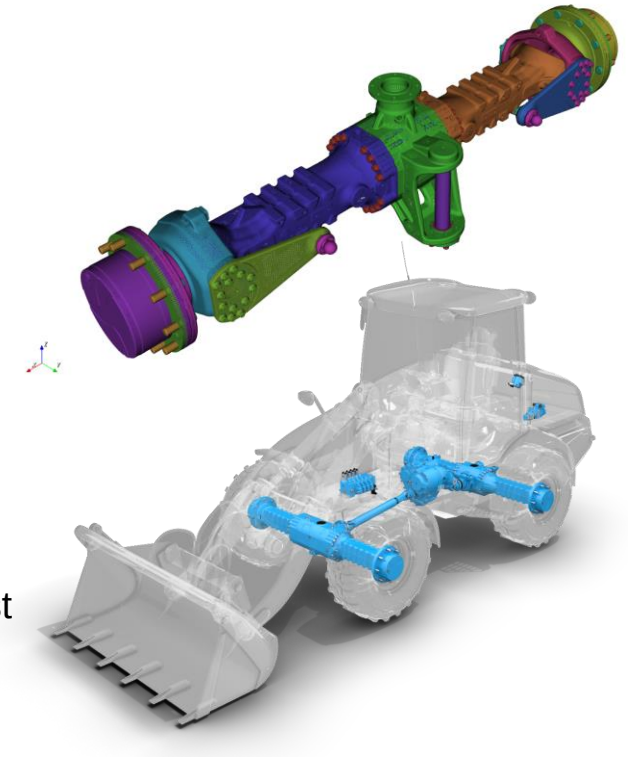
- Incorporate parameters and loads variability in durability design
- Identify loads and locations contributing the most to the damage of the axle structure.
- Determine the best combination of loads and event representative of customer operations

Solution

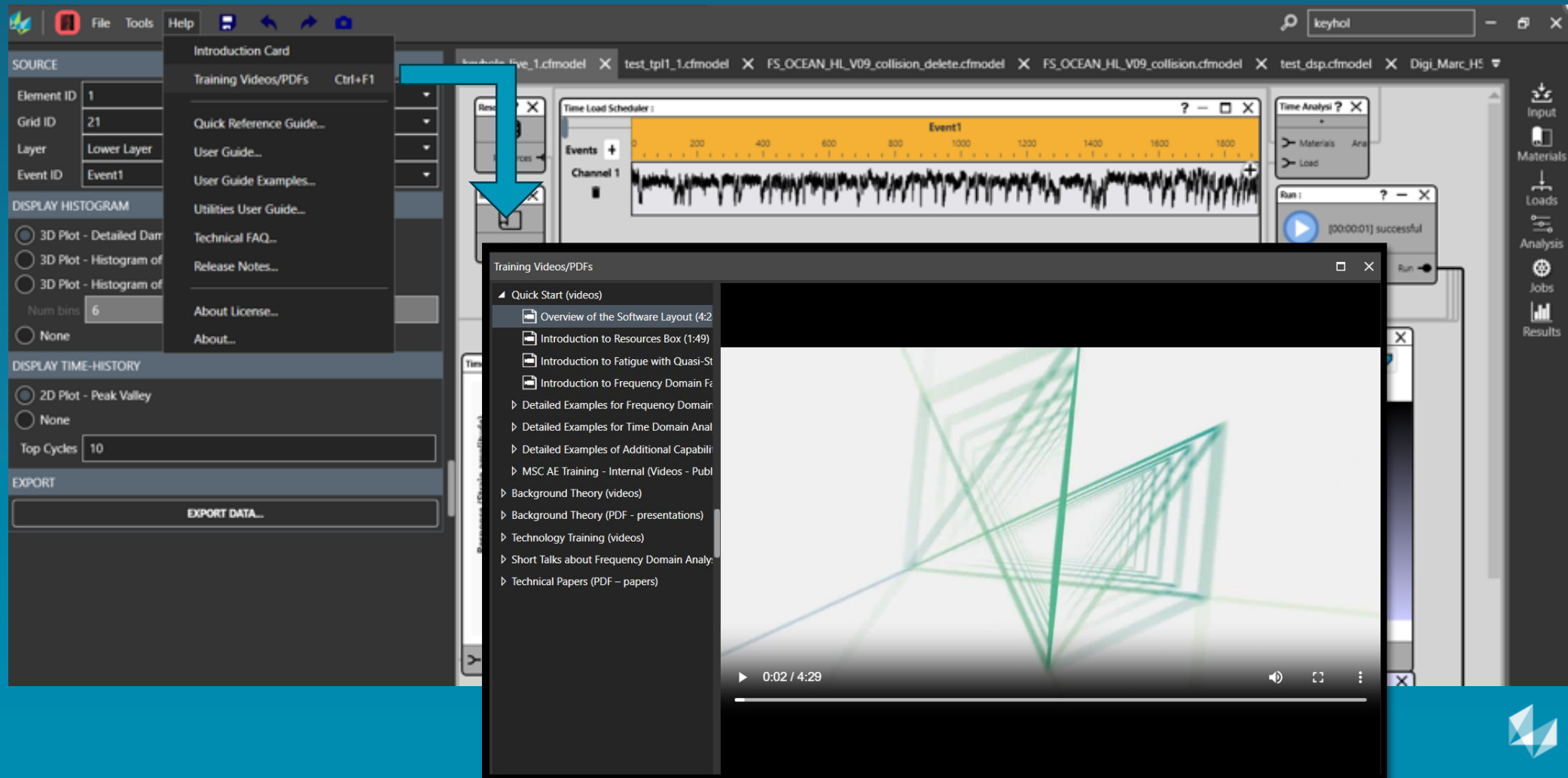
- Robust Design tool provides actionable outputs that quantify the effect of each variable and loading (e.g., y-cycle loading is the most destructive) while ALL input variables vary.
- Surrogate loads through SDI for a most realistic and robust component certification physical test

Benefits

- Gain insight into the underlying interrelations of variables to understand which are the most influential.
- Reduce and improve physical testing.
- Reduce liability costs by exploring the whole design space of loading conditions upfront.



All CAEfatigue (technical) documentation, videos and examples are accessible from the CAEfatigue GUI





File Home Comment View Form Protect Share Help Extras Tell me what you want to do..

Find

CAEfatigue Software

Self Training using **HELP > START**

January 2021
v2021

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Better
Products
Quicker

Time, Budget,
Quality &
Repeatability
important for a
project delivery
organisation

new platforms
– More
complex
engineering,
more levels of
integration

Digital
handshake
between
engineering
toolsets

Digital
handshake
between
engineering and
operations

Product quality
and robustness

Insights into
complex
systems
limited



What differentiates our Durability offering from our competitors?

- We have the **latest and most technologically advanced architecture** combining loads management, fatigue, random response and stochastic simulation software in the market
 - Solver agnostic (Nastran, Marc, Abaqus, Ansys, Optistruct, LS-Dyna)
 - stand-alone GUI, batch control file or FE integrated (Nastran/Marc)
 - multi domain, multi thread, multi tab...
- We deliver these capabilities through **easy-to-use streamlined workflows (templates)** that can be leveraged by durability experts as well as less experienced engineers for **unmatched productivity**
- As a fully Hexagon owned technology, we provide **the most integrated durability and Robust Design solutions**, our **digital handshake** abridging acquired loads, with manufacturing, stress analysis and testing
- Our closely connected teams of **durability simulation experts and solution architects** are super responsive to customer requirements and improvements, always ready to support customers on any of their challenges



HEXAGON

MSC Software

Thank you.



Marco Veltri, PhD
Business Enablement Manager, Multiphysics
Product Management & Business Development

marco.veltri@hexagon.com